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Towards the next development of vitrimers: Recent key topics for the practical application and understanding of the fundamental physics

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ABSTRACT

Vitrimers have emerged as an innovative class of functional cross-linked polymers, providing recyclability, healability, and post-cured malleability without distinct flow. These features are attributed to the relaxation and diffusion of network strands through associative bond exchanges within the network. Significant progress has been made in investigating the chemical library and new functionalities, along with comprehensive studies on fundamental physics, including relaxation and rheological characteristics. Despite a rapid increase in research publications over the past decade, critical challenges remain in practical applications, particularly regarding preparation protocols, control of physical properties, and the development of analytical techniques. Unlike existing reviews focusing on vitrimer design and basic features, this article highlights recent crucial topics, such as vitrimer transformation from commodity polymers, the trade-off between processability and mechanical performance, and the control/analysis of stress relaxation time and topology freezing temperature, and an understanding of rheological properties, based on experimental, simulation, and theoretical studies. The transformation using commodity polymers could introduce a novel upcycling technique. The trade-off issues propose unique vitrimer designs utilizing phase-separated structures and post-molding curing. Moreover, given the strong correlation between relaxation/rheological properties and processability/recyclability/healability, their control and analysis are vital for both foundational physics and practical applications. Throughout the article, we provide insights and pose new open questions for the next development of vitrimer materials.

1. Introduction

Traditionally, a clear distinction exists between thermoplastics and thermosets (Fig. 1, top), determined by the presence or absence of cross-links between polymer chains. The primary difference lies in whether the material is flowable upon heating or soluble when immersed in a solvent; once cross-linking reaction is performed, the material cannot flow and becomes insoluble. Consequently, cross-linking reactions serve as an effective chemical method to enhance thermal stability and solvent resistance. Additionally, mechanical strength is often improved through cross-linking. These benefits were recognized as early as the 20th

Abbreviations: DBU, 1,8-diazabicyclo[5.4.0]undec–7-ene; 2-MI, 2-methylimidazole; AIE, aggregation-induced emission; AFM, atomic force microscope; COOH, carboxylic acid group; CANs, covalent adaptive networks; DSC, differential scanning calorimetry; DIC, digital image correlation; DCB, dynamic covalent bond; FT-IR spectroscopy, Fourier-transform infrared spectroscopy; GMM, generalized Maxwell model; T_g, glass transition temperature; OH, hydroxyl group; KWW function, Kohlrausch-Williams-Watts function; T_m, melting temperature; PDMS, poly(dimethylsiloxane); PHMA, poly(hexylmethacrylate); PMMA, poly(methyl methacrylate); PS, poly(styrene); PB, polybutadiene; PBT, polybutylene terephthalate; PE, polyethylene; PET, polyethylene terephthalate; PI, polyisoprene; PP, polypropylene; RBLM, Renormalized Bond Lifetime Model; SEC, size-exclusion chromatography; SAOS test, small-amplitude oscillatory shear test; SAXS, small-angle X-ray scattering; SB rubber, styrene-butadiene rubber; SBS elastomer, styrene-butadiene-styrene elastomers; SDGs, sustainable development goals; TMA, thermomechanical analysis; TRONs, thermoreversible organic nanofillers; TTS, time-temperature superposition; $Sn(Oct)_2$, tin(II) 2-ethylhexanoate; T_v, topology freezing temperature; WLF equation, Williams-Landel-Ferry equation; Zn(acac)2, zinc acetylacetonate.

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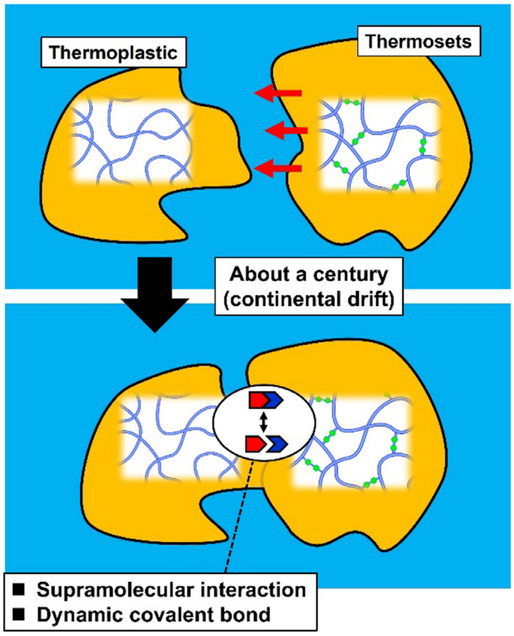


Fig. 1. Schematic illustrating the evolving boundary between thermoplastics and thermosets (cross-linked polymers).

century with the invention of Bakelite, predating the establishment of "makromolekulare Theorie" by Staudinger [1]. Today, a variety of cross-linked polymers have been developed, showcasing enhanced mechanical and thermal properties, aided by advancements in chemical synthetic technology. However, in light of the growing societal emphasis on sustainability, it is crucial to address the drawbacks of cross-linked materials. Their nonflowability and insolubility hinder conventional recycling methods, such as material and chemical recycling, making it difficult to integrate these materials into a sustainable circulation of chemical resources.

Indeed, significant advancements have been made in addressing the aforementioned drawbacks by imparting thermoplastic-like characteristics to cross-linked materials. As a result, the critical distinction between thermoplastics and thermosets (or cross-linked materials) is increasingly blurred from a macroscopic functional perspective (Fig. 1, bottom). For instance, designs utilizing supramolecular interactions have been explored since the 1990s [2–6]. These designs incorporate reversible physical bonds, such as hydrogen bonds, ionic interactions, and coordination bonds, at the cross-link points, allowing the resulting materials to exhibit thermoplastic-like behavior. However, the bond strength of supramolecular interactions is typically much lower than that of covalent bonds [7,8], making supramolecular cross-linked materials vulnerable to solvents, large strains, and repeated deformation. Consequently, it remains challenging to fully replace these weak materials with those constructed via chemical cross-linking. Alternatively, a special type of covalent bond that enables controlled dissociation and reassociation through suitable external stimuli has emerged since the 2000s. These reversible covalent bonds are known as dynamic covalent bonds (DCBs), and cross-linked materials containing DCBs are referred to as covalent adaptive networks (CANs) [9–12]. In the context of conventional CANs, bond exchange occurs in response to stimuli. The reversible bond exchange of DCBs is effectively frozen in the absence of stimuli, preserving potential solvent resistance and mechanical strength. When external stimuli, such as heat, pH changes, or light, are applied, these materials exhibit advantageous sustainable functions, including recyclability and healability.

Importantly, DCBs are currently classified into two major categories [13,14]. The first is the dissociative type, wherein bond exchange occurs through distinct steps of dissociation of the old bond pair and reassociation of the new bond pair. A well-known example of this process is the reversible exchange of Diels-Alder adducts. The second category is the associative type, where bond exchange proceeds via an associated intermediate (Fig. 2a). In this case, the prior association of the new bond pair occurs, followed by the dissociation of the old bond pair during the exchange. Examples of associative-type DCBs include the SN_2 reaction of transesterification and the metathesis reaction of boronic esters. Cross-linked materials utilizing associative-type DCBs are called vitrimers, in which bond exchange occurs through thermally induced chemical reactions within the network. Since the pioneering work by L. Leibler's group in 2011 [15], vitrimers have progressively gained attention due to their unique physical properties and functions. In vitrimer networks, the connectivity of the network strands is maintained throughout the bond exchange due to their associative nature, distinguishing them from dissociative-type CANs. While the network topology is continuously altered during the activated state of bond exchange, the overall cross-link density remains preserved. Beyond the designs based on transesterification and boronic ester metathesis, various network architectures have been explored, incorporating bond exchange mechanisms, such as transcarbamoylation of urethane bonds [16–18], transamination of vinylogous urethane exchanges [19–21], disulfide metathesis [22–24], and olefin metathesis [25–27] with a range of polymer species. It should be noted that some exchange systems involve both dissociative and associative mechanisms, although the underlying mechanisms are not always thoroughly examined. For example, disulfide exchange has been classified into three groups: one is the associative metathesis mechanism, while the other two are dissociative mechanisms mediated by thiyl radicals and thiolate anions. Regarding disulfide exchange, only the first case can be classified as vitrimers in the strict sense. Therefore, the terminology used to describe these materials—whether they are true vitrimers or not—should be applied with caution.

The associative nature of the bond exchange mechanism in vitrimers is reflected in the physical properties. These materials exhibit thermoplastic-like characteristics, yet they differ significantly from conventional thermoplastics [28]. One notable distinction lies in the viscosity-temperature relationship. For amorphous thermoplastics, the viscosity decreases above the glass transition temperature (T_g) according to the Williams-Landel-Ferry (WLF) equation. In contrast, the viscosity of vitrimers begins to decrease above a specific temperature known as the topology freezing temperature (T_v). According to current understanding, vitrimers exhibit a transition at this temperature, T_v, in terms of the viscoelastic state, due to the sufficiently activated bond exchange in the network. Importantly, the viscosity-temperature relationship in vitrimers follows Arrhenius behavior rather than the WLF function, at temperatures above T_v (Fig. 2b). This is attributed to the fact that viscosity changes are primarily governed by the reaction kinetics of bond exchange, which are described by the Arrhenius equation. The Arrhenius dependence of viscosity is well known for vitreous silica; the similarity of viscosity-temperature relationship to vitreous silica is highly unique for vitrimers among organic materials, which is the reason for the term, "vitrimer". In principal, the relaxation time above T_v is also known to adhere to the Arrhenius relationship.

To date, the development of vitrimer designs has been extensively documented in various review articles. These reviews addressed the diversity of network designs utilizing a versatile chemical library, fundamental properties, and the similarities/differences compared to conventional thermoplastics or dissociative CANs [29–33]. Some reviews focused on more specific categories of vitrimer designs, including those utilizing transesterification bond exchange

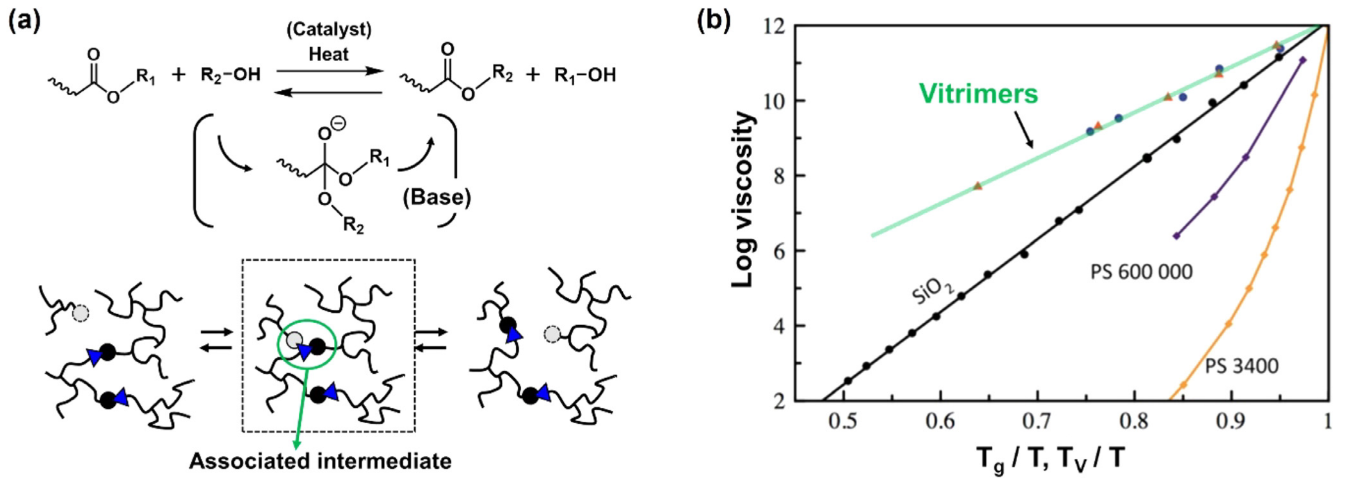


Fig. 2. (a) Schematic representation of associative bond exchange in a vitrimer network, with trans-esterification as an example of this process. (b) Semi-logarithmic plots of viscosity as a function of inverse temperature. Temperatures are normalized by T_v for vitrimers and T_g for other samples. The measured samples include vitrimers with transesterification and vinylogous urethane bond exchange mechanisms, vitreous silica (SiO_2), and poly(styrene) with molecular weights of 600000 (PS 600000) and 3400 (PS 3400). [28], Copyright 2016. Adapted with permission from Royal Society of Chemistry.

[34], transcarbamoylation bond exchange [35,36], bio-based vitrimers [37,38], and composite vitrimers [39,40]. The simulation and theoretical studies for vitrimers was also highlighted as a central topic in recent reviews [41]. Additionally, some reviews highlighted potential applications in advanced areas such as 3D printing [42–44] and liquid crystalline elastomers [45].

The aforementioned reviews are crucial for the practical application of vitrimers. To effectively pursue these applications, it is essential to develop preparation methods that meet the necessary production criteria, ideally focusing on ease and low cost. In this regard, studies have demonstrated the transformation of commodity polymers, such as polyethylene terephthalate (PET) and polybutylene terephthalate (PBT), into vitrimers through a one-shot preparation process. In the first section, we thus summarize the designs for one-shot vitrimer transformation of commodity polymers, the reported mechanisms of transformation, and their significance for realization of sustainable societies. The second section addresses critical challenges that must be overcome for practical applications. Although the Arrhenius viscosity-temperature relationship is a key feature of vitrimers, the gradual viscosity decrease indicated by the Arrhenius relationship can hinder the use of conventional molding techniques, such as injection and extrusion molding. In fact, there is a trade-off between enhancing processability at high temperatures and maintaining mechanical performance at service temperatures. This section illustrates various approaches to tackle these challenges. The relaxation time is another important parameter for characterizing vitrimer materials, as it relates to the efficiency of recycling, healing, and malleability, in addition to viscosity. Controlling and analyzing relaxation time is crucial for regulating these macroscopic functions. In the third section, we discuss influential factors affecting relaxation time, based on the recent studies. The manifestation of certain functions is also influenced by T_v, as strand diffusion and relaxation are activated above this threshold. In the fourth section, we examine conventional estimation methods for T_v, highlighting their defects and errors, as well as alternative approaches for more accurate estimation. Understanding the rheological features across both short and long timescales—from local segmental to large network scales—is another important topic, directly linked to manufacturing and function regulation. In the final sections, we present rheological insights into vitrimers employing small-amplitude oscillatory shear (SAOS) tests. Overall, this review addresses both practically and fundamentally important topics, some of which have been rarely documented in existing literature. Our aim is to provide valuable insights and stimulate new discussions that will contribute to the future development of vitrimers, benefiting both industry and academic researchers.

2. Development of preparation methods

2.1. General preparation manners

The chemical library of vitrimer designs has expanded to encompass a wide range of polymers by incorporating suitable exchangeable units. There are two major preparation strategies. The first involves the curing of multifunctional monomers or oligomers, where polymerization and cross-linking occur simultaneously. A representative example is found in the pioneering reports of vitrimers by Leibler's group [15], where bifunctional epoxy monomers reacted with a mixture of bifunctional and trifunctional carboxylic monomers. The advantage of this method lies in its simplicity; mixing and heating (or UV irradiation) yield the target sample without extensive purification steps. In designs utilizing multiple component monomers, varying the monomer feed ratios allows for the regulation of cross-link density and the fraction of exchangeable units. The second preparation method employs functionalized polymers that contain cross-linkable units at either the end groups or side groups. These component polymers are reacted with cross-linker molecules to form vitrimer networks. The application of controlled living polymerization techniques, such as reversible addition−fragmentation chain-transfer polymerization and atom transfer radical polymerization, facilitates the production of network precursors with precisely regulated chain lengths and distributions of cross-linkable units.

2.2. Transformation from polyesters

In addition to precise network designs, the development of easy and cost-effective preparation methods using commercial polymers is crucial for practical applications [46]. Ideally, the production of highly functionalized vitrimer materials should be achieved without laborious preparation steps. To address this, some studies have proposed a one-shot preparation method for vitrimers, which involves blending the necessary chemicals with commercial polymers and applying heat, all without prior chemical modification of the component polymers.

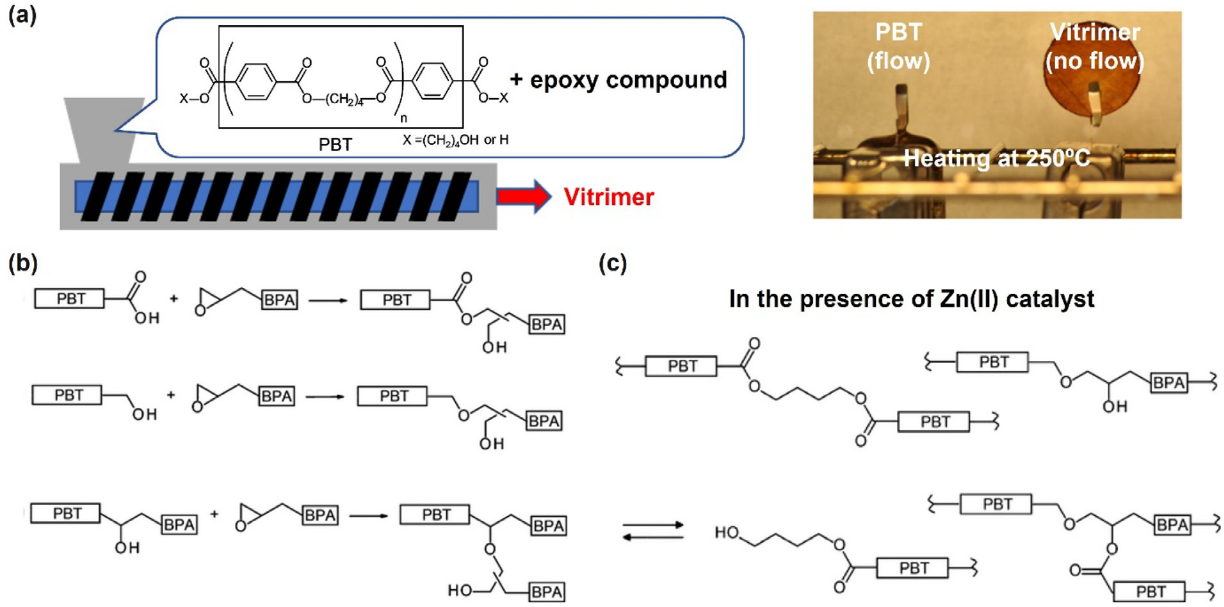


Fig. 3. Vitrimer transformation from PBT. (a) Overview of the vitrimer transformation strategy, including the macroscopic appearances of the original PBT and the resulting vitrimer. (b) Reaction of PBT with diepoxy compounds, both in the absence of catalysts and in the presence of 2-methylimidazole (2-MI). (c) Cross-linking mechanism via transesterification in the presence of zinc acetylacetonate (Zn(acac)$_2$). [47], Copyright 2017. Adapted with permission from American Chemical Society.

As a pioneering study, Demongeot et al. reported the vitrimer transformation of PBT [47]. This transformation was achieved through reactive extrusion, in which PBT was reacted with diepoxy compounds in the presence of catalysts (Fig. 3a). Following this process, the original flowability of PBT was lost after its conversion into vitrimers. The authors investigated the transformation mechanism (i.e., the cross-linking mechanism) by examining the roles of two different catalysts, 2-methylimidazole (2-MI) and zinc acetylacetonate (Zn(acac)$_2$). The original PBT contained carboxylic acid (COOH) or hydroxyl (OH) groups at the chain ends, which were consumed during the reaction with the epoxy groups (Fig. 3b, top and middle), leading to an increase in OH groups due to the ring-opening of the epoxy. In cases of simple mixing of PBT and diepoxy compounds without any catalysts, Fourier-transform infrared (FT-IR) spectroscopy revealed a decrease in COOH groups and an increase in OH groups. However, the axial force detected in the extruder remained low, indicating that some chain extension occurred, but cross-linking did not take place. When 2-MI was used as a catalyst, the rate of change in COOH and OH groups increased significantly, yet the axial force in the extruder did not rise. This suggests that 2-MI facilitated chain extension reactions between the end groups of PBT and the epoxy compounds. In this scenario, branching may have occurred due to reactions between the generated OH groups and epoxy (Fig. 3b, bottom).

In contrast, a significant increase in axial force was observed when Zn(acac)$_2$ was used instead of 2-MI, which ultimately led to the formation of cross-linked materials. This result indicated that transesterification occurred between the ester bonds in PBT and the OH groups generated from the epoxy-opening reaction (Fig. 3c). Additionally, it was noted that the increase in axial force followed the consumption of a large fraction of the initially present COOH groups. The final materials exhibited a rubbery plateau above the melting temperature (T_m), with the concentration of epoxy compounds in the mixture affecting the plateau modulus and, consequently, the cross-link density. Despite the cross-linked structure, stress relaxation and Arrhenius dependence of relaxation time were observed, confirming the incorporation of vitrimer characteristics. In a subsequent study based on this design [48], the authors proposed a two-step transformation process to produce vitrimers from PBT, aimed at avoiding screw blocking during reactive processing. This involved a prior preparation of a pre-vitrimer mixture through short-time extrusion, followed by heating the prepared mixture for solid-state reactions. The transformation from pre-vitrimers to final vitrimers was supported by rheology tests.

Similarly, Kimura and Hayashi reported a one-shot vitrimer transformation using commercial amorphous polyesters that have chemical structure similar to PET [49,50]. In their study, they incorporated tetra-functionalized epoxy compounds 4,4′-Methylenebis(*N, N*-diglycidylaniline), along with a base catalyst 1,8-diazabicyclo[5.4.0]undec–7-ene (4-Epoxy and DBU in Fig. 4a, respectively). Mixture samples were prepared via either solution casting or twin-screw extrusion, and subsequent heating of the dried samples resulted in the formation of vitrimers. The appearance of a stable rubbery plateau confirmed cross-linking following the transformation (Fig. 4b), while stress relaxation and other vitrimer functions were observed due to transesterification bond exchange in the cross-linked samples. As anticipated, the feed concentration of epoxy compounds influenced the transformation rate (i.e., the cross-linking rate) and the overall cross-link density. Notably, an increase in epoxy concentration did not always correlate with enhanced transformation rates and cross-link density; the authors identified an optimal feed concentration for such enhancements.

The transformation mechanism was primarily assessed using FT-IR, size-exclusion chromatography (SEC), and rheology tests. The interpretation of the mechanism was consistent with previous findings from Demongeot et al. [47,48], involving reactions between epoxy and COOH or OH end groups, as well as transformations between ester bonds and the OH groups generated from epoxy ring-opening. Additionally, they observed that the polyesters decomposed, likely via hydrolysis, during solution casting or mixing in the extruder. This decomposition produced reactive end groups (i.e., COOH and OH groups) in the fragmented chains, enhancing the likelihood of reactions with epoxy molecules. Thus, the overall mechanism comprised three steps: the initial decomposition of polyesters, chain extension through epoxy reactions, and transesterification between the chains (Fig. 4c). Notably, the initial decomposition of polyesters was also observed in the study by Wu

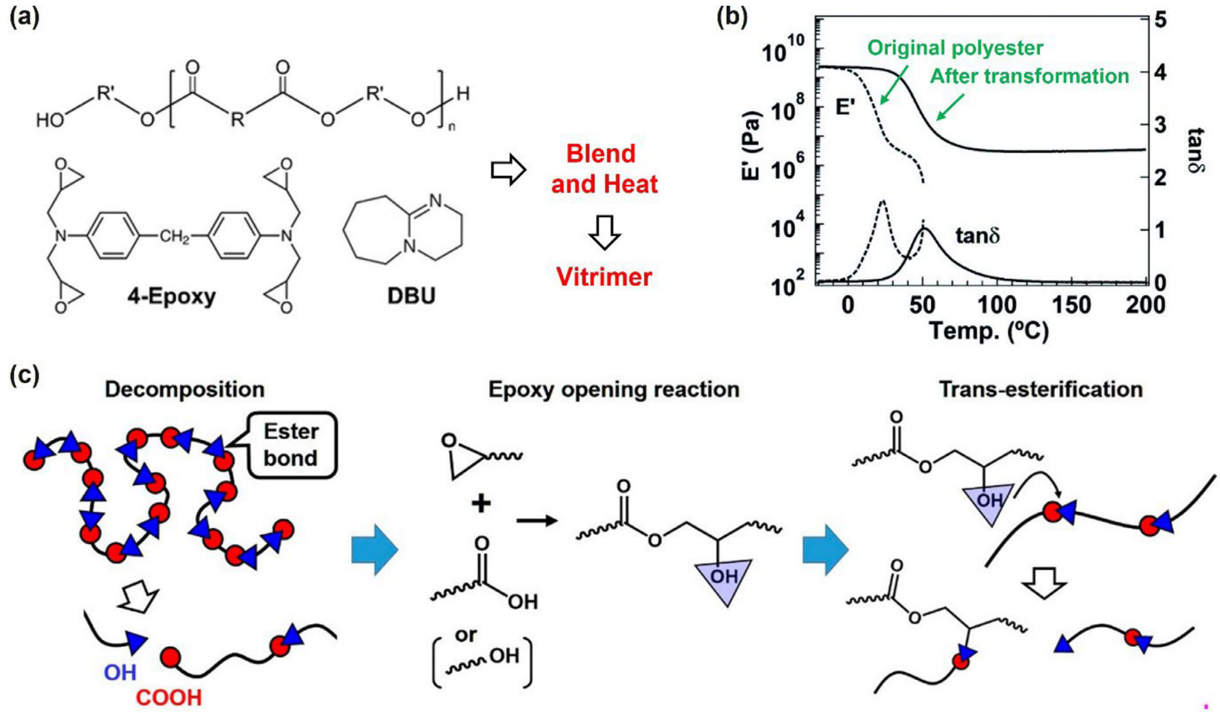


Fig. 4. Vitrimer transformation from amorphous polyesters. (a) Molecular design. (b) Temperature-sweep rheology data for the original polyester and the sample after the transformation. [49], Copyright 2022. Adapted with permission from Royal Society of Chemistry. (c) Schematic representation of the transformation mechanism. [50], Copyright 2023. Adapted with permission from American Chemical Society.

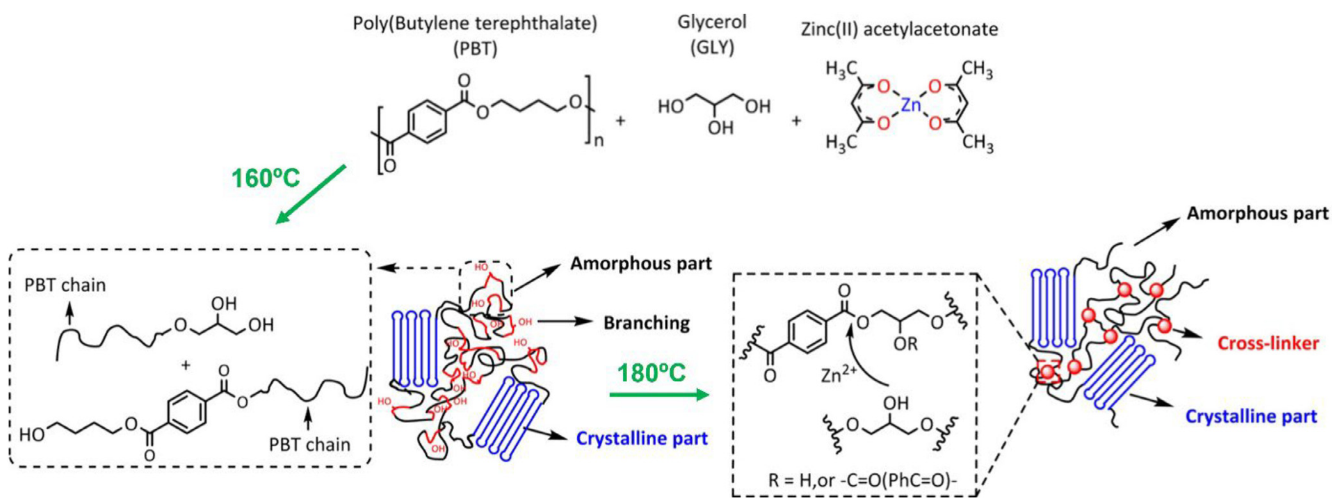


Fig. 5. Vitrimer transformation from PBT. The molecular design and schematic representation of the transformation mechanism are illustrated. [52], Copyright 2017. Adapted with permission from American Chemical Society.

et al., who investigated the vitrimer transformation of PET using the same tetra-functional epoxy compound, 4,4′-methylenebis(*N*, *N*-diglycidylaniline) [51]. They further demonstrated that the tertiary amine in this compound served as an internal catalyst for cross-linking and bond exchange via transesterification within the final network.

Multi-functionalized alcohol compounds can also serve as alternatives to epoxy compounds. For example, Zhou et al. reported the vitrimer transformation of polybutylene terephthalate (PBT) via a solid-state reaction, using glycerol, a simple tetra-functional alcohol compound, as the cross-linker and $Zn(acac)_2$ as the catalyst (Fig. 5) [52]. Heating the mixture at 160 °C resulted in a decrease in molecular weight and an increase in molecular weight distribution, indicating the alcoholysis of PBT chains by glycerol. Raising the temperature to 180 °C facilitated condensation reactions between the OH and ester end groups generated by alcoholysis, promoting the recovery of molecular weight. Prolonged heating at 180 °C led to transesterification between the pendant OH groups and the ester bonds in the chains, resulting in the formation of a cross-linked network. Since the temperature of 180 °C was below T_m of PBT, transesterification was interpreted to occur in the amorphous phase. Following cross-linking, a clear rubbery plateau was observed above T_m, and the plateau modulus increased with higher initial glycerol concentrations. The vitrimer characteristics of the cross-linked samples were evidenced by stress relaxation, where bond exchange occurred through transesterification facilitated by the $Zn(acac)_2$ catalyst. In subsequent studies, the authors investigated the effects of catalyst amounts on thermal, rheological,

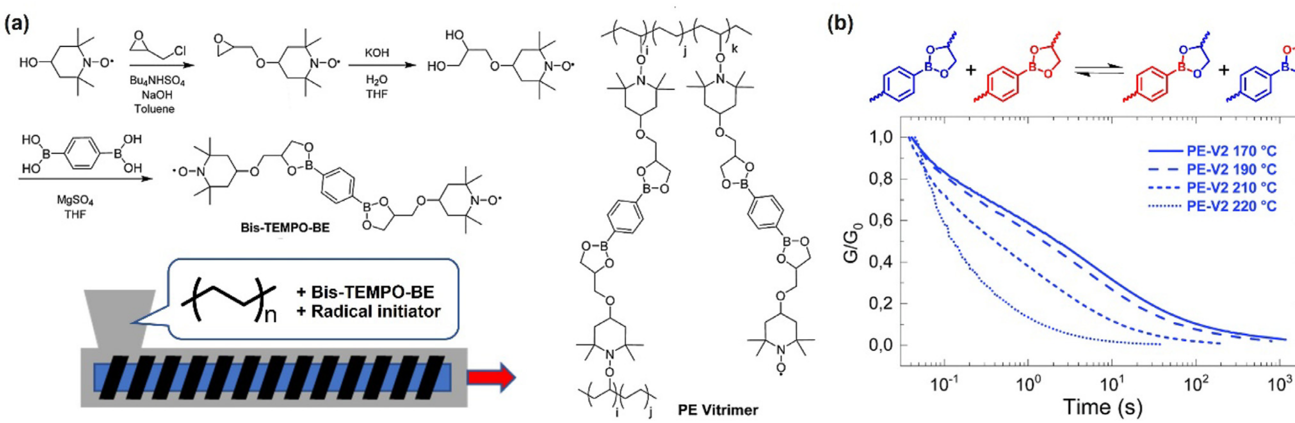


Fig. 6. Vitrimer transformation from polyethylene. (a) Molecular design. (b) Stress relaxation data at various temperatures for the sample after the transformation. The bond exchange mechanism of boronic ester metathesis is also illustrated. [58], Copyright 2019. Adapted with permission from Royal Society of Chemistry.

and relaxation properties [53]. As a combined strategy, Qiu et al. demonstrated the vitrimer transformation of polyethylene terephthalate (PET) using both epoxy and alcohol compounds as cross-linking agents [54]. The varying feed ratios of these compounds allowed for the regulation of the network structure and associated physical properties.

Such one-shot transformations into vitrimers are not limited to PET and PBT; they can also be applied to other commercial polyesters, such as poly(butylene succinate) [55] and poly(butylene adipate-co-terephthalate) [56], using appropriate cross-linking agents and catalysts. These starting polyesters are biobased and/or potentially biodegradable, adding significant value to the studies.

2.3. Transformation from other commodity polymers

The preparation of vitrimers from polyolefins, such as polyethylene (PE) and polypropylene (PP), is also significant, given that these are among the most produced commodity polymers. A common strategy for this transformation involves cross-linking through radical-based reactions [32,57–59]. For example, Caffy and Nicolaÿ reported the preparation of vitrimers from PE by incorporating associative exchangeable cross-links via nitroxide radical coupling (Fig. 6a) [58]. The bond exchange occurred through boronic ester metathesis in the final network, resulting in complete stress relaxation (Fig. 6b). As an alternative strategy, Png et al. demonstrated the vitrimer transformation of PE and PP using a carbene C–H insertion approach, incorporating imine-based exchangeable units at the cross-link points [60]. Their study also applied this strategy to polystyrene (PS) and poly(methyl methacrylate) (PMMA), as well as to mixtures of PE and PS. In addition to these advanced chemical techniques, some studies have shown that vitrimers can be prepared using partially modified PE or PP with epoxy or maleic anhydride units [61–64], which are commercially available. These modified polymers readily reacted with epoxy cross-linkers, leading to the formation of transesterification vitrimers.

Polydienes contain abundant double bonds along their chains, making them reactive under milder conditions compared to the main chains of PE or PP, thus facilitating one-shot vitrimer transformations. Breuillac et al. reported several one-shot preparations of vitrimers using dioxaborolane metathesis (Fig. 7a), where cross-linking reactions were conducted with polybutadiene (PB) or polyisoprene (PI) [65,66]. Given that conventional vulcanized rubbers are highly challenging to recycle [67,68], vitrimer-type polydiene rubbers are promising alternatives for development. Additionally, styrene-butadiene rubbers (SBR) or styrene-butadiene-styrene elastomers (SBS elastomer) possess reactive components from the butadiene units, allowing for various one-shot vitrimer transformations [69–71]. Notably, the study by Fang et al. compared the relaxation and creep properties of SBS elastomer-based vitrimers and SBR-based vitrimers (block vitrimer and random vitrimer in Fig. 7b) [71]. They found that the phase-separated structure in the block vitrimer exhibited greater creep resistance at high temperatures compared to the random vitrimer (Fig. 7c). Similar to the cases of PE and PP, partially modified polydienes with epoxy or maleic anhydride units are commercially available and can be utilized for vitrimer preparation [72–74]. Vitrimer transformations from Nylon-6 [75] and ethylene-vinyl acetate copolymer serve as additional commercially important examples [76]. As such, strategies for vitrimer transformation from commodity polymers are expanding, with precise control of the transformation rate emerging as a critical issue for true practical applications.

2.4. Another meaning of vitrimer transformation from commodity polymers

The significance of vitrimer transformations from commodity polymers lies not only in the ease and cost-effectiveness of production but also in the potential for developing advanced upcycling techniques of waste polymers. A major barrier to recycling is the infrastructure required for the collection of waste products, while the cost of recycling remains another critical issue. The recycling process involves several steps, including collection, washing, and remolding with high heat application, which often results in the cost of producing recycled materials being higher than that of virgin materials. Additionally, recycled materials typically exhibit inferior mechanical and thermal properties compared to virgin ones, often due to thermal or UV degradation [77,78]. PET bottles, for example, have a high collection rate in many countries; however, most collected PET is incinerated to reclaim the energy content of the plastic, primarily due to cost and deterioration concerns [79]. Incineration of used polymeric products results in various problems, including the loss of chemical resources and emissions of CO_2 and other harmful gases [80,81]. To realize effective recycling systems within a capitalist framework, it is essential to develop upcycling systems that transform original products into higher-value items, rather than relying on horizontal or cascade recycling [82,83].

The aforementioned vitrimer transformations from commodity polymers can be considered upcycling techniques when waste polymers are used as starting materials. For instance, transform-

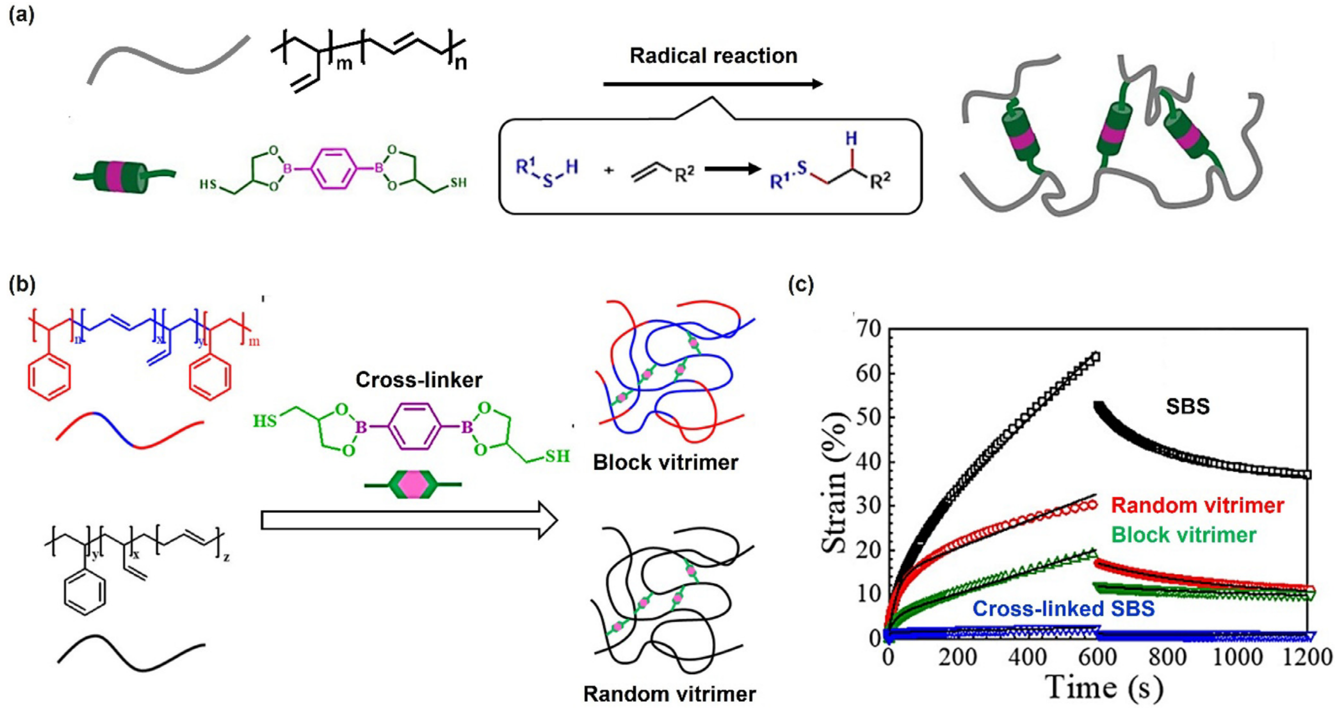


Fig. 7. (a) Molecular design of the transformation from polybutadiene. [65], Copyright 2019. Adapted with permission from American Chemical Society. (b) Molecular design of the transformation from SBS. (c) Isothermal creep data at 150 °C for SBS, Random vitrimer, Block vitrimer, and chemically cross-linked SBS (refer to the text for sample details). [71], Copyright 2022. Adapted with permission from American Chemical Society.

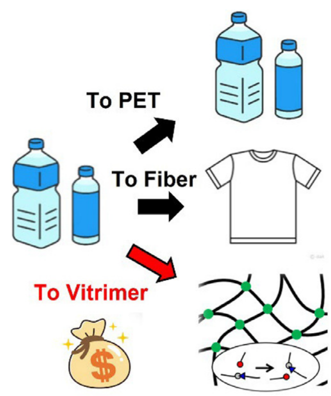


Fig. 8. Schematic illustrating horizontal recycling (i.e., PET to PET), cascade recycling (i.e., PET to fiber), and upgraded recycling (upcycling) into vitrimers.

ing waste PET into vitrimer PET enhances mechanical and thermal properties due to network formation while incorporating additional functions of vitrimers (Fig. 8). The development of upcycling systems could bolster business motivation, which is necessary for achieving truly sustainable circular systems. Since the establishment of sustainable development goals (SDGs) in 2015, various countries and regions have set concrete targets, including increasing the use of recycled polymers. In this context, further development of vitrimer transformations from commodity polymers holds significant value from a global perspective.

3. Critical problems for production and recycle processes

3.1. Trade-off between the mechanical performance and processability

In addition to the preparation protocols, several factors need to be solved for the practical application of vitrimers. A significant issue is the trade-off between mechanical properties and processability. To improve processability, it is essential to develop systems with short relaxation time or low viscosity. For this purpose, utilizing low T_g polymers or networks with low cross-link density may be advantageous. However, in these cases, the mechanical properties—such as modulus, fracture stress, and creep resistance—at service temperatures may not be adequately anticipated.

To address this essential trade-off relationship, Nicolaÿ's group conducted several valuable studies. One of their studies demonstrated the preparation of elastomeric vitrimers through the reactive processing of immiscible high-T_g and low-T_g component polymers, resulting in phase-separated networks with domain sizes ranging from several hundred nanometers to a few microns (Fig. 9a) [84]. Importantly, the exchangeable cross-link connections involving boronic ester moieties were located exclusively at the interphase of the domains. Consequently, most polymer segments, except for the chains near the domain interface, remained free from cross-linking, facilitating easy flow above the activation temperature for bond exchange. At service temperatures (~30 °C), the phase-separated samples containing glassy phases ($Ext_{<75\ \mu m}$ and $Ext_{>75\ \mu m}$ in Fig. 9b where the subscript "<75 μm" or ">75 μm" indicates the size of high-T_g polymer particles after grinding) exhibited significantly superior mechanical properties compared to the homogeneously cross-linked elastomeric samples lacking glassy components (BisBE Vit in Fig. 9b). The flow properties at elevated temperatures were assessed using rheological tests. Despite the enhanced mechanical strength, the flowability of the phase-separated samples was superior, as indicated by the observation of a cross-over regime in SAOS data, where the loss modulus (G'') exceeded the storage modulus (G') at high temperatures (Fig. 9c for $Ext_{>75\ \mu m}$). The authors attributed this behavior to sufficient diffusion and relaxation, enabled by the activation of bond exchange at the domain interphase. In contrast, the homogeneously cross-linked elastomeric samples without glassy components did not exhibit a cross-over regime within the same frequency or temperature range, as all the chains remained intact with surrounding chains through the associative cross-links.

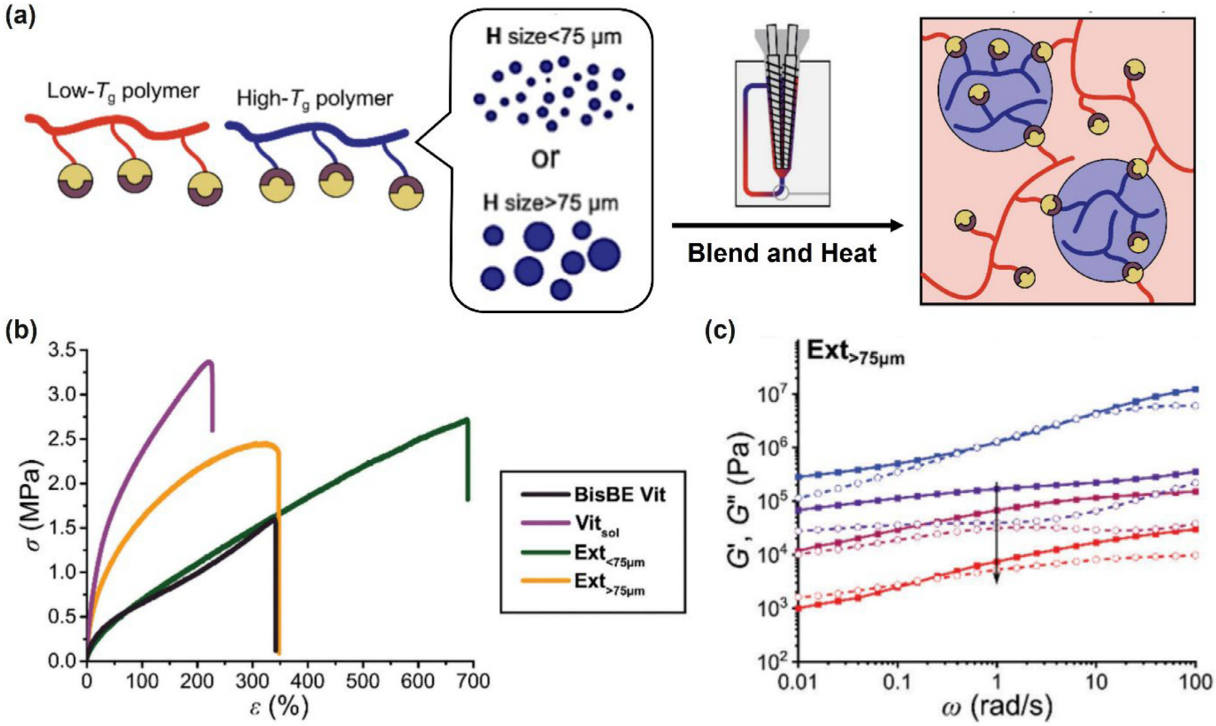


Fig. 9. Strategy to overcome the trade-off between mechanical performance and processability. (a) Design of blending low T_g and high T_g polymers to create phase-separated vitrimer networks. (b) Stress–strain curves. (c) Frequency-sweep rheology data for the phase-separated vitrimer sample. [84], Copyright 2023. Adapted with permission from John Wiley & Sons Inc.

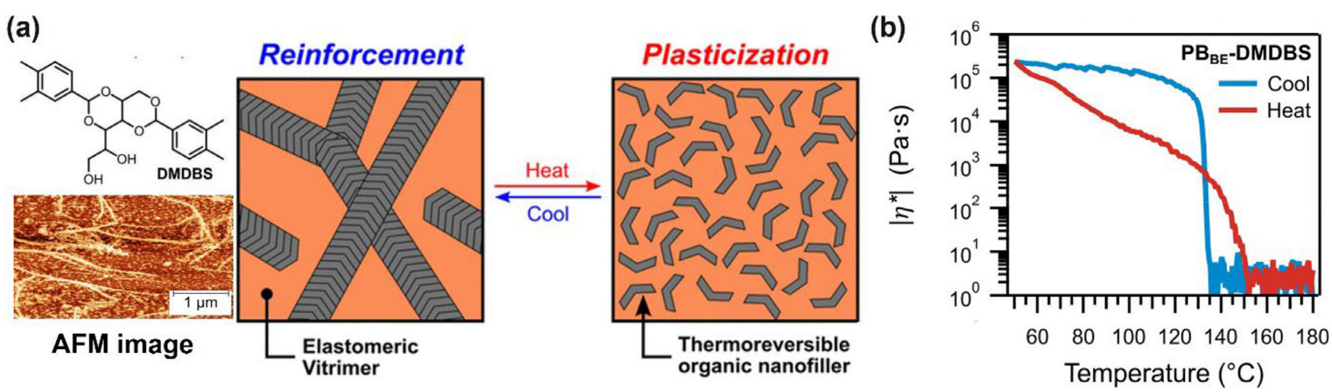


Fig. 10. (a) Design utilizing thermoreversible organic nanofillers (TRONs). (b) Temperature dependence of complex viscosity for the blend of network precursor polymers with TRONs. [85], Copyright 2024. Adapted with permission from American Chemical Society.

In the following study, the authors proposed an alternative approach to address the trade-off problem by utilizing special fillers [85]. The key innovation was the use of thermoreversible organic nanofillers (TRONs), specifically 1,3:2,4-bis(3,4-dimethylbenzylidene)sorbitol (DMDBS) (Fig. 10a). A notable feature of DMDBS is its fiber-shaped aggregation, which occurs through a combination of π–π stacking and hydrogen bonding at low temperatures. At higher temperatures, weakening of the supramolecular interactions leads to the collapse of this aggregation. When TRONs are incorporated into the vitrimer matrix, the aggregation acts as reinforcing fillers. Conversely, in the disaggregated state, TRONs function as plasticizers. Fig. 10b illustrates the change in complex viscosity as a function of temperature for a mixture of polybutadiene precursors with DMDBS at a weight fraction of 1.5 wt%. A sudden change in viscosity was observed in the temperature range of 130–150 °C during both heating and cooling, attributed to the aggregation and disaggregation of DMDBS, although some hysteresis was noted. Fiber-shaped aggregation was indeed observed in the cross-linked vitrimer composite containing DMDBS, as shown in the atomic force microscope (AFM) image. This sample exhibited simultaneous enhancements in tensile properties and relaxation rate, compared to the vitrimer sample without DMDBS, as designed. Thus, these advanced concepts present an impressive solution to the potential trade-off problem between mechanical performance and processability.

3.2. Essential problem for manufacturing and recycling process

While a solution to the aforementioned trade-off problem is highly desirable, the fundamental challenge in achieving easy processing lies in developing methods to attain low viscosity values. The difficulty in achieving low viscosity stems from a key characteristic of vitrimers. For typical vitrimers, the viscosity follows Arrhenius dependence at temperatures sufficiently high for

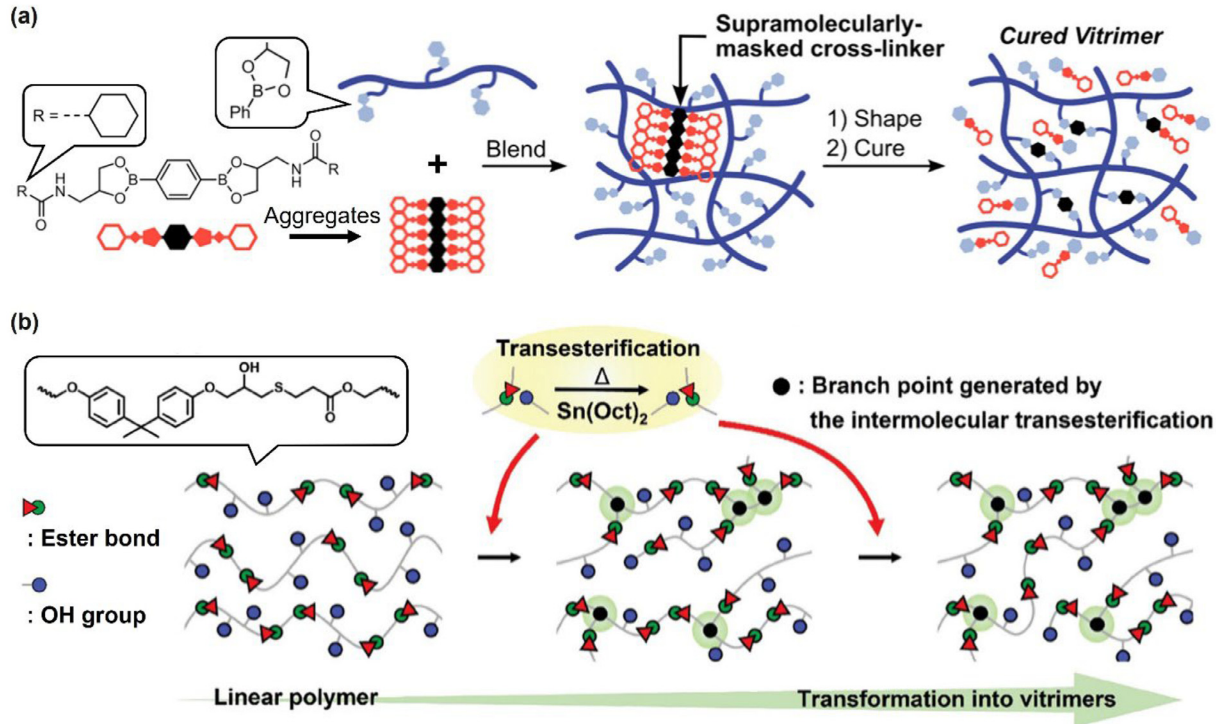


Fig. 11. (a) Post-molding curing strategies utilizing masked cross-linkers. [99], Copyright 2024. Adapted with permission from American Chemical Society. (b) Transformation of polyesters with abundant OH side groups into vitrimers. [100], Copyright 2024. Adapted with permission from John Wiley & Sons, Inc.

bond exchange activation. Thus, the temperature dependence of viscosity is much weaker than that for thermoplastics. Note that some reports indicate Arrhenius behavior has also been observed in dissociative bond exchange systems [14,86]. These dissociative systems, which exhibit vitrimer-like behaviors, are specifically referred to as vitrimer-like materials to distinguish them from true vitrimers that operate via associative bond exchange mechanisms [87–94]. In both true vitrimers and vitrimer-like materials, the weak temperature dependence of viscosity provides several advantages in applications. For instance, malleability is a well-known function of vitrimers, wherein macroscopic flow does not occur during high-temperature processing, unlike in thermoplastics. In adhesive applications, maintaining high viscosity during heating inherently prevents over-melting, thereby enhancing dimensional stability [95].

However, the high viscosity of vitrimers inhibits the applicability of conventional molding techniques, such as injection molding and extrusion, posing a critical challenge for the manufacture of vitrimer products with desired shapes. In practice, few studies have demonstrated the use of injection or extrusion molding for cured vitrimers [46,96]. Notably, dissociative CANs generally exhibit a stronger temperature-viscosity relationship (i.e., a more pronounced reduction in viscosity with increasing temperature) in the bond exchange state, enabling the application of conventional extrusion techniques for cured materials [97,98]. This behavior arises because the network connectivity collapses during bond exchange in dissociative systems, resulting in thermoplastic-like flow. To address this issue for vitrimers, some studies have proposed post-molding curing strategies with highly controlled cross-linking progression. As an interesting example, Formon et al. reported a design for controlled cross-linking using special cross-linkers with cross-linkable units masked by supramolecular interactions (Fig. 11a) [99]. The cross-linker featured amide groups with hydrogen bonding capabilities, promoting aggregation and phase separation. The precursor polymers and cross-linkers were blended and molded with heating in the extruder, while the cross-linkers remained inactive in this state. For the extruded blend, a higher temperature was applied to unmask the cross-linkers by breaking the hydrogen bond interactions, thus enabling post-molding curing. Another study demonstrated a transformation of thermoplastic linear polyesters into vitrimer via intermolecular transesterification [100], where the starting polyester contained abundant OH side groups along the polyester chain (Fig. 11b). Thus, this system could be applied for post-molding curing. Indeed, the vitrimer transformation from commodity polymers using reactive extrusion or solid-state reactions, as described in Section 2, could serve as an option for post-molding curing.

The aforementioned strategies may facilitate the manufacturing of initial products; however, the viscosity issue persists for cured vitrimers. Conventional recycling processes typically employ injection or extrusion molding, making it challenging for high-viscosity vitrimers to be reprocessed into recycled products with desired shapes. While compression molding could be applicable, it is limited to producing only film-shaped products. A solution to this critical limitation is essential for fully leveraging the recyclability of vitrimers in practical systems. The high viscosity problem is also related to interfacial adhesion in composite vitrimers. Although some studies have demonstrated the fabrication of multilayer composites from a single layer of fiber-reinforced composite vitrimers [22,101,102], where good interfacial adhesion (or welding) was observed, interfacial adhesion remains a critical factor to prevent delamination. The high viscosity, resulting from limited chain mobility and diffusion, impedes the interpenetration of chains between layers, which is an essential consideration for the practical application of composite-type vitrimers.

4. Control of relaxation properties

4.1. General knowledge from chemical perspective

Vitrimers exhibit stress relaxation due to bond exchange within the network. In principle, the relaxation time follows an Arrhenius

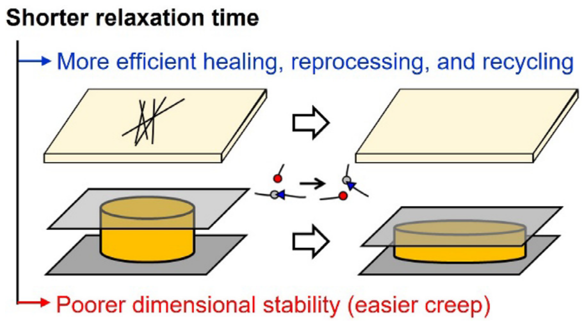


Fig. 12. Schematic illustrating the advantages and disadvantages of vitrimers with a shorter relaxation time.

dependence, as repeatedly mentioned. While some studies discuss relaxation properties based on data from frequency sweep rheology, stress relaxation tests are predominantly used to characterize vitrimer properties. Research has shown that the relaxation time is linked to the efficiency of recyclability and healability for vitrimers or classical CANs. As illustrated at the top of Fig. 12, materials with shorter relaxation times can be healed, reprocessed, and recycled at lower temperatures and within shorter processing durations [103–105]. On the other hand, samples with shorter relaxation times exhibit issues such as poorer dimensional stability and undesired creep during bond exchange, as illustrated at the bottom of Fig. 12. Overall, it is essential to develop methods for controlling relaxation time and to understand the factors influencing the relaxation time. An additional significance of this understanding lies in the fact that relaxation time is directly proportional to viscosity, which subsequently affects processability.

To date, various influential factors affecting the relaxation time have been explored. A well-known example involves catalytic activation systems, where the type and amount of catalysts are essential for controlling relaxation time [106–110]. It should be noted that the presence of specific chemical units— which catalytically works or stabilizes the transition state— influences the rate of bond exchange reactions or, in some cases, the reaction mechanism itself, which eventually affects or usually enhanced the relaxation rate. These are indeed the basic principles of internal catalysis and neighboring group participation [111–122]. Additionally, the bond exchange rate within the network is influenced by the polarity of the polymer strands [123,124], similar to what is observed in solution reactions.

The aforementioned examples for controlling relaxation time are predominantly based on the chemical aspects of bond exchange reactions. Since bond exchange efficiency in the network correlates with the diffusion of strands or exchangeable units seeking partners, various network parameters—including cross-link density [71,125,126], concentration of bond exchangeable units [127,128], entanglements [92,129], T_g of the network polymers [130–132], network defects [133,134] and the presence of phase-separation [135–139]—are known to impact relaxation properties. Beyond these topics, the effects of network heterogeneity—specifically the location of exchangeable units and the random formation of permanent bonds—have been investigated through the design of suitable networks [140–142]. These studies are particularly important in practical applications, as such heterogeneities may arise during large-scale production.

4.2. The paradox of cross-link density and exchangeable units fraction effects

The cross-link density within the network governs strand mobility and, consequently, affects the diffusivity of the exchangeable units. To evaluate this effect, Soman and Evans designed a network with controlled cross-link density by reacting OH-terminated telechelic short polyethylene with boric acid cross-linkers, facilitating bond exchange via boronic transesterification (Fig. 13a) [143]. The alkyl length of the polyethylene strands was varied from C_4 to C_{12} to tune the cross-link density, which was confirmed by the rubbery plateau modulus observed in rheological measurements (Fig. 13b). Fig. 13c presents the frequency sweep rheology data, showing that the relaxation time (τ)—determined by the intersection of G' and G'' — was longer for samples with shorter alkyl lengths (indicating higher cross-link density). A similar result was obtained from stress relaxation tests. The simple interpretation is that an increase in cross-link density results in greater restriction of strand mobility, limiting the diffusion required to find exchange partners. This tendency has also been reported by other research groups [144,145].

In contrast, some studies have reported completely opposite results regarding cross-link density, based on similar designs using telechelic strand polymers and cross-linkers. Chen et al. developed a transesterification network by reacting telechelic short polyethylene with end epoxy groups and tri-functional carboxylic acid cross-linkers (Fig. 14a) [146]. The molecular weight of the strand polymer was varied from 200 to 640 g/mol to regulate the cross-link density. Fig. 14b presents plots of the natural log of τ from stress relaxation tests as a function of temperature. The increase in the molecular weight of the strands resulted in longer τ, which contrasts with the findings of Soman and Evans described above [143]. In their interpretation, the distance between bond-exchangeable units is shorter for networks with higher cross-link density, leading to more frequent bond exchange reactions. Similar results with the same interpretation have been observed in studies from other research groups [123,147], indicating that this is not an isolated case.

These results indicate that the effects of cross-link density manifest in different ways. To address this contradiction, a clear explanation is provided by the simulation study conducted by Zhao et al., based on coarse-grained molecular dynamics simulations [148]. In the modeled network, an increase in cross-link density was accompanied by an increase in the concentration of exchangeable units (Fig. 15a), which aligns with the aforementioned experimental studies. Furthermore, they varied the activation energy for bond swapping (ΔE_{SW}), allowing the results to be applicable to various vitrimer systems with different bond exchange energies. Fig. 15b illustrates the relationship between dynamic cross-link density ρ and bond exchange lifetime (τ_{SW}) in the network; ρ and τ_{SW} is correlated with the concentration of the bond exchange unit and the relaxation time, respectively. As ρ increased, τ_{SW} initially decreased and then increased, resulting in a minimum τ_{SW} observed at a certain ρ. The authors interpreted this finding by noting that bond exchange kinetics are primarily influenced by two opposing effects: on one hand, increased cross-link density restricts strand mobility, inhibiting bond exchanges; on the other hand, higher dynamic bond density enhances the frequency of encounters among bond-exchangeable units. Consequently, beyond a certain density (i.e., $\rho = 25$–45 % as shown in the figure), the restrictive effect of the network outweighs the facilitating effect of cross-linking, leading to a slower bond exchange rate at higher ρ. Based on this study, the opposite trends observed in the previously mentioned experimental studies can be understood.

On a different note, the above study by Zhao et al. investigated the effects of applied force on the bond exchange rate [148]. In Fig. 15c, the bond exchange autocorrelation function (C_{SW}), which correlates with the relaxation and diffusion rates, was compared between the equilibrated and non-equilibrated networks under stretching. The faster decay of the bond exchange autocorrelation function in the non-equilibrated stretched state indicated

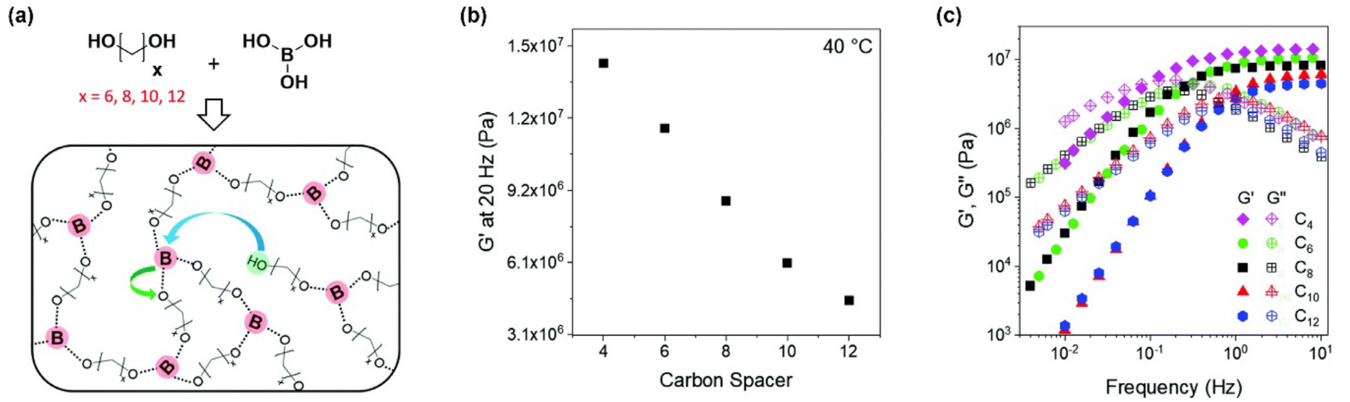
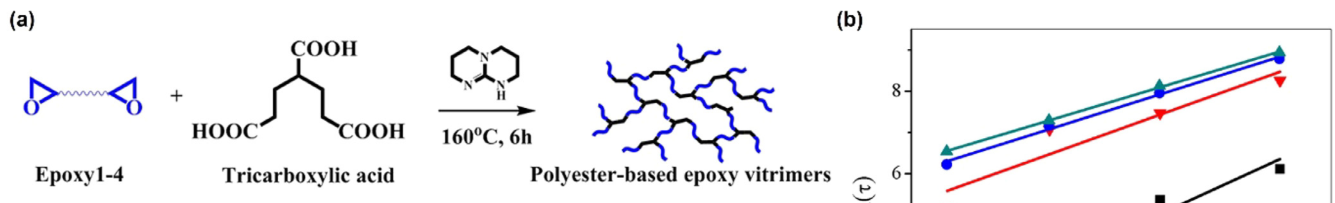


Fig. 13. (a) Molecular design to investigate the effects of cross-link density. (b) Plots of G' at 20 Hz measured at 40 °C (rubbery modulus) as a function of carbon numbers in the spacer. (c) Frequency sweep rheology data for vitrimer samples with varying carbon numbers in the spacer. [143], Copyright 2021. Adapted with permission from Royal Society of Chemistry.



Epoxy monomers	Molecular weight	Names of epoxy vitrimers
Epoxy 1	~202g/mol	E_{202}-vitrimer
Epoxy 2	~380g/mol	E_{380}-vitrimer
Epoxy 3	~500g/mol	E_{500}-vitrimer
Epoxy 4	~640g/mol	E_{640}-vitrimer

Fig. 14. (a) Molecular design to investigate the effects of cross-link density. (b) Semi-logarithmic plots of relaxation time (τ) as a function of inverse temperature for samples with varying strand lengths. [146], Copyright 2021. Adapted with permission from American Chemical Society.

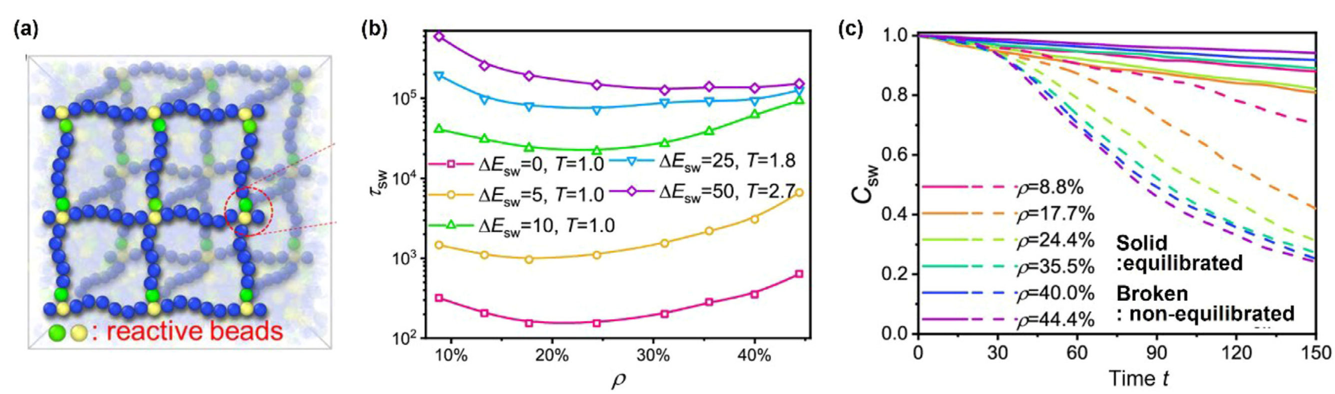


Fig. 15. (a) Schematic of the coarse-grained model of the cross-link network with distributed reactive beads. Each polymer chain contains one type of reactive bead (green or yellow), and the exchangeable network is formed by the reaction between green and yellow beads. (b) Plots of bond exchange lifetime (τ_{SW}) as a function of dynamic cross-link density (ρ) for systems with varying potential activation energies for the bond swapping (ΔE_{SW}). (c) Difference of the bond exchange autocorrelation function (C_{SW}) for equilibrated and non-equilibrated network with varying ρ. [148], Copyright 2024. Adapted with permission from American Chemical Society.

more rapid bond exchange. This result is attributed to the applied force, which modifies the energy landscape for bond dynamics, as supported by other theoretical studies [149,150]. Additionally, some experimental studies have demonstrated the effects of strain on the relaxation time for vitrimers and other transient networks, showing that shorter relaxation times are observed at higher strains [151–153]. These results imply that viscosity is also influenced by the strain field, providing important insights for the processing of vitrimers [154].

The examples discussed above treated the systems in which both the cross-link density and the concentration of bond-exchangeable units were simultaneously varied. In these cases, isolating the pure effects of cross-link density is difficult. Only a few studies have designed networks where the concentration of exchangeable units was kept constant, allowing cross-link density to be the sole variable parameter. However, even in these cases, the effects of cross-link density manifested in conflicting ways. In one instance, an increase in cross-link density shortened the relaxation time [155], while in another case, the opposite result was observed [134,145]. These contradictory findings arise from the interplay of two competing factors: in samples with higher cross-link density, the restriction of strand mobility is greater, while the fraction of strained strands is also larger. The former leads to slower relaxation, whereas the latter results in faster relaxation. To clarify these conflicting results, precise simulations or theoretical analyses as well as more experimental studies should be conducted.

As previously discussed, the fraction of exchangeable units is a critical parameter in determining relaxation behavior. Similar to the effects of cross-link density, the influence of the fraction of exchangeable units should be examined while holding other influential parameters constant. Some studies have attempted to isolate the pure effects of the exchangeable unit fraction by maintaining the same cross-link density across samples for comparison [156,157]. As expected, these results demonstrated that samples with a higher fraction of exchangeable units exhibited faster relaxation. However, the impact on the activation energy for bond exchange (E_a) warrants careful consideration. Usually, E_a is estimated from Arrhenius plots of relaxation time. In an ideal case, E_a should remain unaffected by the fraction of exchangeable units within the network, based on fundamental principles of chemical reactions, which posit that the activation energy is independent of reactive unit concentration. Such an ideal scenario was reported for a study of transesterification vitrimers where the cross-link density, catalyst concentration, and T_g were nearly identical among samples [157]. Nonetheless, various studies have shown that the E_a derived from Arrhenius plots of relaxation time is often significantly higher than values estimated from model solution reactions using small molecules [130,158,159]. This suggests that the E_a obtained from vitrimer relaxation data includes contributions from the diffusion of strands or exchangeable units constrained within the cross-linked network, as well as segmental motion [125,160]. Therefore, small variations in the position and distribution of exchangeable units can influence E_a, even when the apparent cross-link density is consistent across samples. Further experimental or simulation studies are necessary to develop a comprehensive understanding of how bond-exchange unit concentration affects E_a, particularly regarding the estimation of the energy barrier for strand diffusion. This topic will be revisited in Section 6, focusing on rheological analysis.

4.3. Interpretation and analysis of stress relaxation data

Finally, we provide some notes on the analysis of stress relaxation data. In most articles on vitrimers, relaxation time is defined as the time at which the relaxation modulus decays to 1/e of the initial modulus, assuming Maxwell relaxation characterized by a single relaxation time (Fig. 16a, top). However, studies of observing a single Maxwell relaxation are rare. Instead, other studies have utilized fitting based on the stretched exponential function, known as the Kohlrausch-Williams-Watts (KWW) function (Fig. 16a, middle), which assumes a distribution of relaxation times [91,161]. In the KWW function, the value of β indicates the distribution breadth of relaxation times; β equals 1 corresponds to standard Maxwell model behavior, while a lower β signifies a broader relaxation mode distribution. Although the KWW function is a phenomenological model, stress relaxation curves of vitrimers are often well-fitted by this function. This suggests that there is a distribution of relaxation times within the network, even though each bond exchange reaction has a specific time constant.

Simulation studies have yielded significant insights into the distribution of relaxation times. Lin et al. examined the time dependence of the bond number autocorrelation function, which serves as an indicator of the time decay of the fraction of original bond pairs and is thus utilized as an alternative to the stress relaxation curve [162]. By fitting the KWW functions to the normalized curve, they observed that the value of β decreased, indicating a widening of the distribution, as the temperature decreased or approached T_g (Fig. 16b). Furthermore, they explored the effects of the energy barrier for bond exchange (λ) on the value of β; they found that β increased for systems with a larger λ, signifying a narrower distribution of relaxation time. In another study, Zhao et al. performed a similar simulation analysis based on the bond number autocorrelation function [163]. Their fitting of the KWW functions to the normalized curve revealed that the value of β increased (Fig. 16c), indicating a narrowing of the distribution, as the potential barrier for bond swapping (ΔE_{SW}) increased, corresponding to cases of slower bond exchange. These results from Lin et al. and Zhao et al. indicates that bond exchange kinetics can serve as a significant rate-limiting factor, outweighing contributions from segmental mobility. Paradoxically, these findings suggest that segmental motion and diffusion factors must contribute to the relaxation properties of vitrimers. Furthermore, increases in cross-link density and the incorporation of permanent cross-links were found to broaden the distribution of relaxation times, according to both experimental and simulation studies [145,162,164,165]. In some cases, fitting using two-term KWW functions has been performed, assuming distinct slow and fast relaxation modes. Such examples can be found in block copolymer systems and composite vitrimers [166,167]; in the former, each block exhibits a distinctly different relaxation time, while in the latter, the bound rubber phase is assumed to have a different relaxation time.

As an alternative evaluation of the relaxation behavior, the generalized Maxwell model (GMM) may be fit to rheology data (Fig. 16a, bottom) [161]. With the GMM, one can obtain a discrete relaxation spectrum, which contains information on the number of existing relaxation modes, along with their contributions and corresponding relaxation times. A continuous relaxation spectrum may be obtained by replacing the summation in the GMM with an integral. Although the modes estimated from the GMM are not necessarily unique, they describe the strength and timescales of the various relaxation mechanisms within a given system, providing an interpretation that is complementary to KWW fitting. Statistical analysis even may be performed on the discrete or continuous relaxation spectrum to obtain a terminal relaxation time [161]. Some studies have utilized relaxation spectra derived from this analysis to describe the distributed stress relaxation behaviors in vitrimers [168–170]. The appropriate application of the GMM and discrete/continuous relaxation spectra allows for extraction of the effects of measurement conditions (i.e., temperature and strain) and molecular parameters, such as the presence of entanglements and permanent cross-links.

5. Evaluation of topology-freezing temperature

5.1. Conventional methods to determine t_v

In general, vitrimer materials undergo a transition between the viscoelastic solid state and the malleable viscoelastic liquid state through the activation and deactivation of bond exchange within the network. T_v, the topology freezing temperature, is termed for the onset temperature for this transition. Conceptually, the material begins to creep, and relaxation becomes apparent around T_v. Thus, T_v is regarded as the temperature at which the activation of bond exchange starts to be reflected in macroscopic properties. It is important to note that measuring the exact activation temperature for bond exchange at the chemical bond scale is challenging, as typical associative bond exchange reactions, such as SN_2 and metathesis reactions, do not show a decrease or increase in the fraction of exchangeable units. Most probably, T_v may not correspond to the exact onset temperature of the chemical reaction

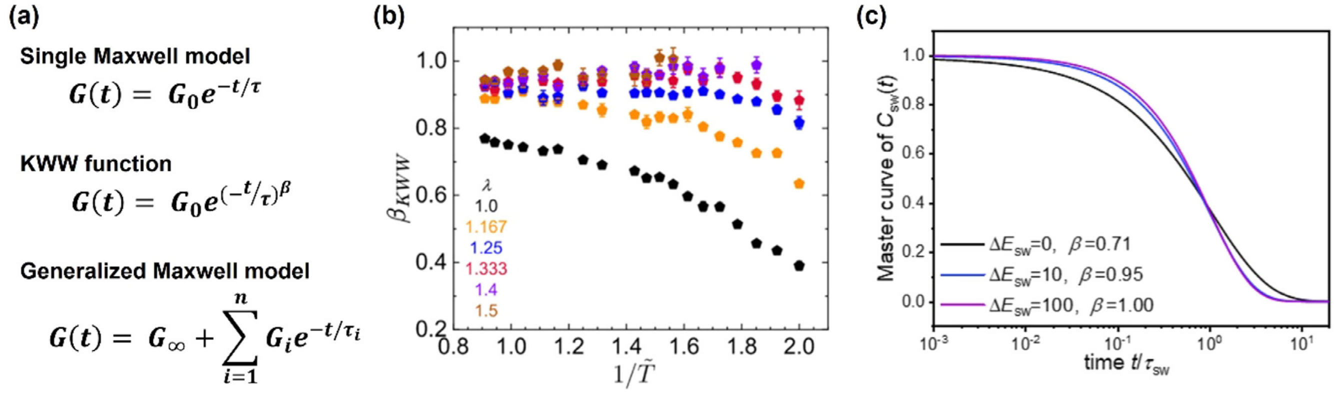


Fig. 16. (a) Representative functions describing the stress relaxation curves for vitrimers. (b) Temperature dependence of β values obtained from fitting the bond number autocorrelation function with KWW functions. In the figure, λ serves as an indicator of the energy barrier for bond exchange. [162], Copyright 2025. Adapted with permission from American Chemical Society. (c) Curves of the bond number autocorrelation function (C_{SW}) and β values from the KWW fitting for systems with different potential barriers in bond swapping (ΔE_{SW}). [163], Copyright 2023. Adapted with permission from American Chemical Society.

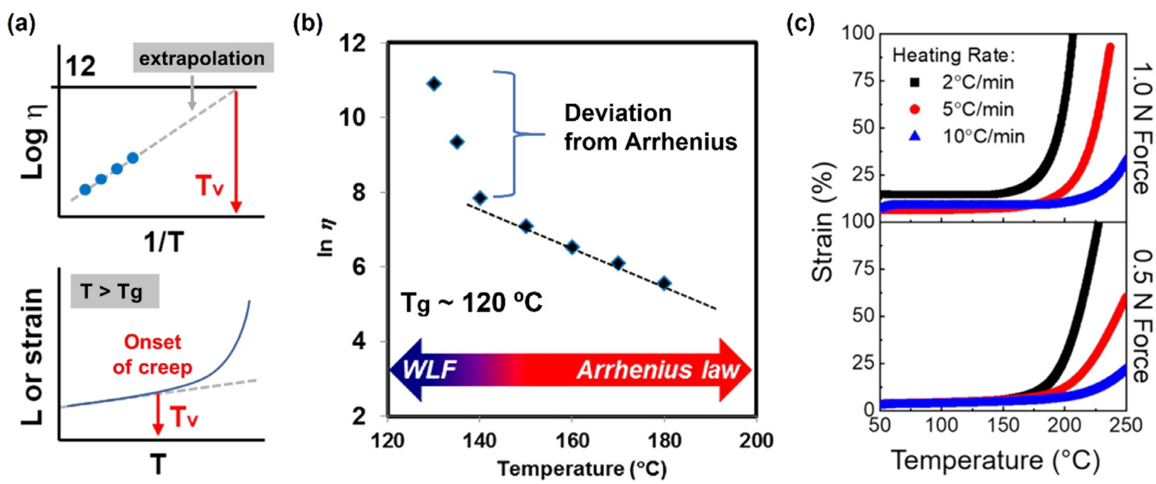


Fig. 17. (a) Schematic for estimating T_v from temperature-viscosity plots and temperature-ramp creep tests. (b) Semi-logarithmic plots of viscosity as a function of temperature near T_g. [114], Copyright 2017. Adapted with permission from American Chemical Society. (c) Effects of heating rate and tensile force in temperature-ramp creep for estimating T_v. [171], Copyright 2021. Adapted with permission from American Chemical Society.

within the network. As mentioned in the previous section, the reaction kinetics derived from small molecule tests are not directly applicable to the vitrimer network. Consequently, determining the precise activation temperature of the reaction itself, as well as understanding its difference from T_v within the network, remains an important open question.

Nevertheless, various methods have been developed to estimate T_v for different vitrimer samples; representative methods rely on the temperature dependence of viscosity and creep (or expansion) behaviors (Fig. 17a). Similar to relaxation time, the values of T_v are influenced by the type and amount of catalyst used in catalytic systems [106,109,171], as well as network parameters such as cross-link density and the concentration of bond-exchangeable units [127,143,145,147].

In one representative estimation, T_v is defined as the temperature at which the viscosity reaches 10^{12} Pa·s (Fig. 17a, top). To determine T_v, the Arrhenius plots of viscosity are constructed, and extrapolation to lower temperatures provides the value of T_v corresponding to a viscosity of 10^{12} Pa·s. Instead of using measured viscosity values, T_v can also be estimated from plots of "calculated" viscosity, utilizing values of relaxation time τ and modulus G, where viscosity η is calculated based on the Maxwell model definition, $\eta = G \times \tau$. Strictly speaking, this calculation requires the sample to exhibit terminal flow, which can be confirmed by ensuring zero-shear viscosity in frequency-sweep rheology tests or steady-state behavior in isothermal creep tests. In any case, the extrapolation assumes a constant dependence without accounting for deviations from Arrhenius behavior, such as those caused by the glass transition. In fact, some studies have shown that the Arrhenius dependence of η or τ observed at temperatures significantly higher than T_g does not hold near T_g (Fig. 17b) [114,172]. Consequently, the extrapolated T_v value estimated by this method should be considered hypothetical.

Another representative estimation of T_v is based on temperature-ramp creep (Fig. 17a, bottom); T_v is defined as the onset of creep at which sudden softening (i.e., deviation from constant length change) is observed. Thermomechanical analysis (TMA) is typically conducted in tension mode for this estimation; however, recent report has demonstrated that measurements in compression mode can also be suitable [173]. In compression mode, dimensional changes are detected during the heating process, with the response shifting from an increase due to expansion to a decrease due to softening, providing a clear peak for T_v. Regardless of the testing mode, these estimations offer T_v values without assumptions, unlike those derived from the temperature dependence of viscosity or relaxation time. Although these estimations appear more straightforward, several critical issues must be noted. First, similar to T_g investigations by differential scanning

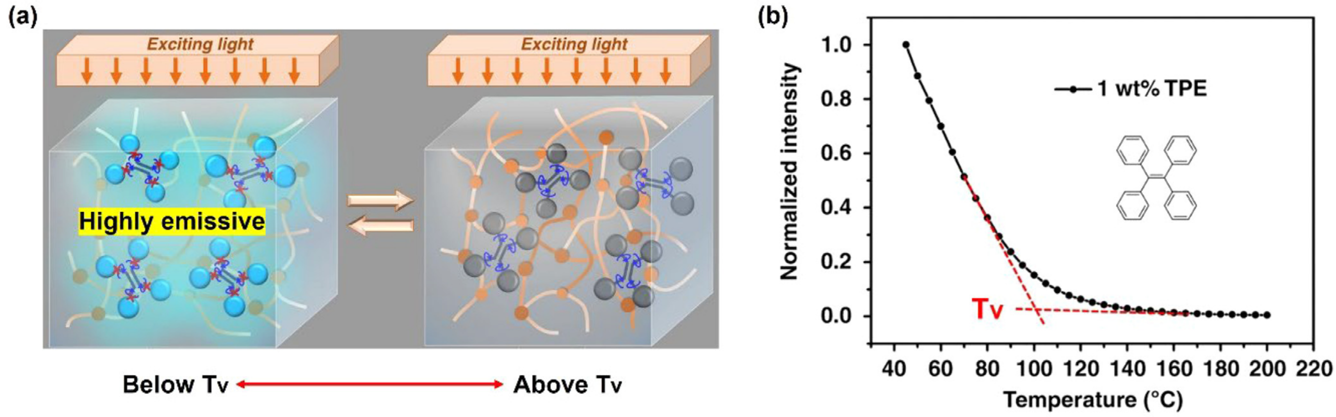


Fig. 18. (a) Schematic illustrating the highly emissive state below T_v and the weakly emissive state above T_v in AIE-doped vitrimers. (b) Temperature dependence of fluorescence intensity for 1 wt% TPE-doped epoxy vitrimer, with intensity normalized to 45 °C. [176], Copyright 2019. Adapted with permission from Springer Nature.

calorimetry (DSC), the heating rate affects the values of T_v. Second, the applied force also influences T_v values. Fig. 17c illustrates the dependencies of heating rates (2, 5, and 10 °C/min) and applied forces (0.5 N and 1 N) for transesterification vitrimers, according to the study by Hubbard et al. [171], which used temperature-ramp creep in tensile mode. A faster heating rate resulted in a higher T_v, whereas a larger applied force tended to yield a lower T_v. A more precise investigation of the force dependence of T_v was reported by Kaiser et al. [174], who developed a method to define the onset of creep based on derivative analysis. In an extreme case, the authors conducted force-free dilatometry combined with digital image correlation (DIC), where no significant thermal expansion was detected at temperatures near the T_v pre-estimated by creep tests under tensile force. Similarly, Klingler et al. investigated dimensional changes in a stress-free state using modulation TMA for transesterification vitrimers, concluding that net thermal expansion across T_v was inconspicuous [175].

Therefore, the conventional methods mentioned above face challenges in accurately estimating the absolute value of T_v, although T_v can serve as an indicator of processing (or reprocessing) temperature and the upper limit of service temperature before critical plastic deformation occurs. Nevertheless, these methods allow for straightforward estimation of T_v, which can facilitate rough comparisons between samples when measurements are performed under consistent and fair conditions.

5.2. Determination of T_v from stress-free methods

Based on the aforementioned knowledge, the estimation of T_v should ideally be conducted using static and force-free methods. To meet this criterion, Yang et al. proposed a method for estimating T_v under static conditions using aggregation-induced emission (AIE) luminogens as fluorescent probes (Fig. 18a) [176]. This study represents a pioneering effort in the stress-free detection of T_v. AIE molecules are unique organic compounds known for their excellent emission properties in aggregated states. The aggregation of AIE molecules restricts their intramolecular motion, leading to intense fluorescence emission, which has previously been utilized to detect T_g in uncross-linked polymer [177]. In their study, the authors blended 1 wt% of tetraphenylethene (TPE) as a fluorescent probe into transesterification vitrimers. As representative data, Fig. 18b presents the fluorescence intensity plotted as a function of temperature, with the Y-axis representing fluorescence intensities normalized to the value at 45 °C. During heating, the intensity gradually decreased and eventually reached a constant minimum above 120 °C. The intersection of two tangent lines was assigned to T_v, based on the assumption that below T_v, chain mobility in the network is highly restricted due to the freezing of bond exchange. In this state, AIE molecules are intramolecularly associated and therefore highly emissive. As the temperature approaches T_v, this intramolecular association weakens. Above T_v, sufficient activation of bond exchange enhances molecular mobility, subsequently reducing fluorescence emission. It is important to note that the variation in fluorescence emission is thermally reversible, unlike the irreversible plastic deformation observed in creep tests. Moreover, when T_g is present within the measured temperature range, the inflection in intensity plots appears at two temperatures, indicating a stepwise increase in molecular mobility across both T_g and T_v. This finding provides significant insight into the existence of two distinct rubbery states for amorphous vitrimers, where molecular mobility differs markedly below and above T_v.

As another example, the estimation of T_v under stress-free static conditions was reported by Arbe et al. using small-angle X-ray scattering (SAXS) [178]. They conducted temperature-ramp SAXS on self-assembled imine exchange vitrimers, where variations in distance between the self-assembled clusters enabled the estimation of T_v. Specifically, the sample was prepared by cross-linking bis-aldehyde-terminated polyisoprene (PI) with triamine cross-linkers (Fig. 19a). The imine cross-link units self-assembled into nanoclusters, with bond exchange occurring via dynamic imine exchange within these clusters. Fig. 19b presents selected SAXS spectra during heating, illustrating the changes in the positions of peak tops. Fig. 19c summarizes the temperature dependence of the intercluster correlation distance D. The light blue and red lines represent thermal expansion in the glassy state and rubbery state, respectively, as estimated from the variation of interchain distance with temperature below and above T_g. In the low-temperature range, the variation of D below T_g resembled thermal expansion. A clear increase in D was observed at T_g (ca. 220 K), with a steeper change detected at higher temperatures (orange line). The authors attributed the inflection point to an additional transition associated with bond exchange, suggesting an increase in the free volume available for the clusters. In the rubbery state, the plot reveals two distinct linear regimes separated by a transition region marked by the shaded area. From these data, the transition occurring around 260–280 °C was assigned as T_v. To support this tentative assignment, T_v was also estimated using another method based on the temperature dependence of zero-shear viscosity, according to the conventional definition with $\eta = 10^{12}$ Pa·s. The T_v estimated from the viscosity plots (272 ± 14 K) was consistent with the T_v determined from SAXS, within experimental error, validating the accuracy of the method based on the variation of D. Thus, determining T_v from two different perspectives is recommended to confirm the accuracy of each individual estimation.

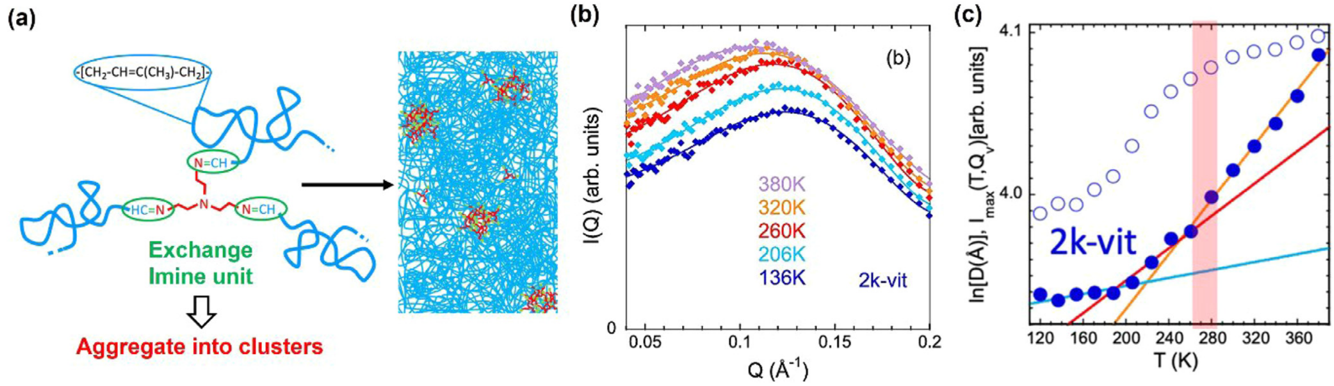


Fig. 19. (a) Schematic of a phase-separated vitrimer network. (b) SAXS data at various temperatures. (c) Temperature dependence of the logarithmic inter-cluster distance ln D (filled blue circles) and maximum intensity (I_{max}) (empty blue circles). The lines in different colors and the red shaded area are explained in the text. [178], Copyright 2023. Adapted with permission from American Chemical Society.

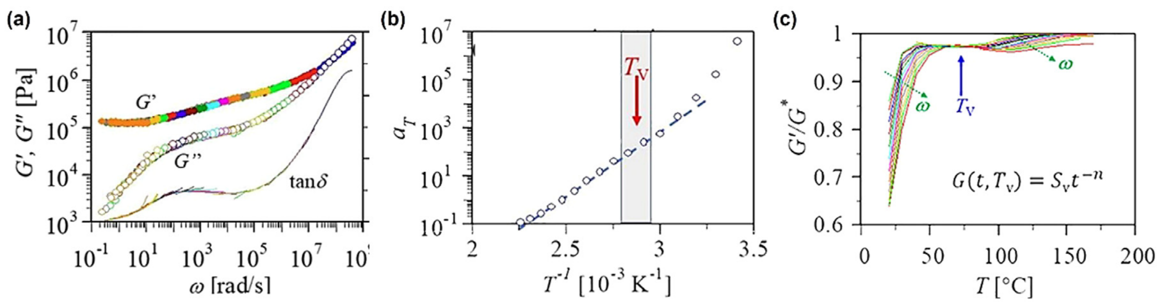


Fig. 20. Master curves for SAOS of transesterification vitrimers. (b) Temperature dependence of shift factors (a_T) in creating the master curve. (c) Temperature dependence of normalized storage modulus (G'/G^*) measured at different frequencies. [181], Copyright 2020. Adapted with permission from American Chemical Society.

5.3. Determination of T_v from SAOS measurements

Recently, rheological analyses based on SAOS tests have been employed to estimate T_v [179,180]. Some reports have investigated the temperature dependence of shift factors a_T in creating master curves from frequency sweeps, leading to the determination of T_v. It is important to note that the applicability of time-temperature superposition for vitrimer dynamics is a significant topic, which will be discussed in the next section. For example, Fang et al. explored various aspects of the rheological features of transesterification vitrimers [181]. In their plots of the temperature dependence of a_T derived from frequency sweep data, they observed an Arrhenius dependence of a_T in the high-temperature region (Fig. 20a and b), with the onset temperature of this dependence assigned to T_v. Another interesting finding emerged from the multi-frequency temperature-sweep data. They noted that several rheological quantities, including the loss tangent (tan δ), loss angle ($2\delta/\pi$), normalized storage modulus (G'/G^*), and normalized loss modulus (G''/G^*), all became independent of frequency at a certain small temperature range (Fig. 20c). This distinct temperature or temperature range was also assigned to T_v, which was consistent with the T_v determined from the a_T plots.

Although extrapolating WLF dependence to the high-temperature region usually leads to a quasi-Arrhenius dependence, other experimental and simulation vitrimer studies have revealed a clear transition of a_T from Williams-Landel-Ferry (WLF) dependence in the low-temperature region to Arrhenius dependence in the high-temperature region; the transition temperature is considered to be T_v. In these cases, the temperature dependence of a_T in the Arrhenius region cannot be accurately represented by extrapolating the WLF dependence. At low temperatures ($< T_v$), a_T follows WLF dependence, where the rheology of the vitrimer network is dominated by the segmental dynamics of strand chains. The interpretation of the Arrhenius dependence of a_T can be summarized as follows; in vitrimers, at sufficiently high temperatures for bond exchange, the relaxation time τ or viscosity η exhibits Arrhenius dependence with temperature. Given the fundamental relationship between a_T and τ or η, it is reasonable to expect that a_T should follow Arrhenius dependence. Thus, the transition from WLF to Arrhenius behavior at a certain temperature provides a solid rationale for assigning T_v to this transition temperature.

As mentioned in the previous section, determining T_v from two different perspectives is advantageous. Hayashi et al. assessed the accuracy of T_v assigned from the plots of a_T in comparison to T_v derived from structural analysis for phase-separated vitrimer-like materials [182]. The sample was prepared using poly(ethyl acrylate) copolymers bearing pyridine side groups randomly distributed along the chain, which were cross-linked by 1,6-diiodohexane via a pyridine quaternization reaction (Fig. 21a). The quaternized pyridine underwent exchange via trans-*N*-alkylation at high temperatures, resulting in vitrimer-like properties. Another noteworthy feature was the formation of nano-domains of quaternized pyridine units, arising from the repulsive forces against the poly(acrylate) strands. Previous studies have demonstrated that bond exchange progresses in an inter-domain manner [92,93]. For this phase-separated system, temperature-ramp SAXS provided insights into the temperature dependence of the distance between the domains (d). The values of d were estimated from fitting the SAXS spectra based on the Yarusso-Cooper model [183], which were then plotted as a function of temperature in Fig. 21b. The initial increase in d at approximately 20 °C was attributed to the glass transition, above which a constant increase in d was observed. A steeper change occurred at temperatures above 130 °C, indicating a shift in the temperature dependence of free volume within the network. The authors interpreted this onset of the steeper change as indicative of bond exchange activation, assigning this onset temperature as the structural T_v. The rheological T_v was subsequently estimated from the plots of a_T, with a deviation from WLF depen-

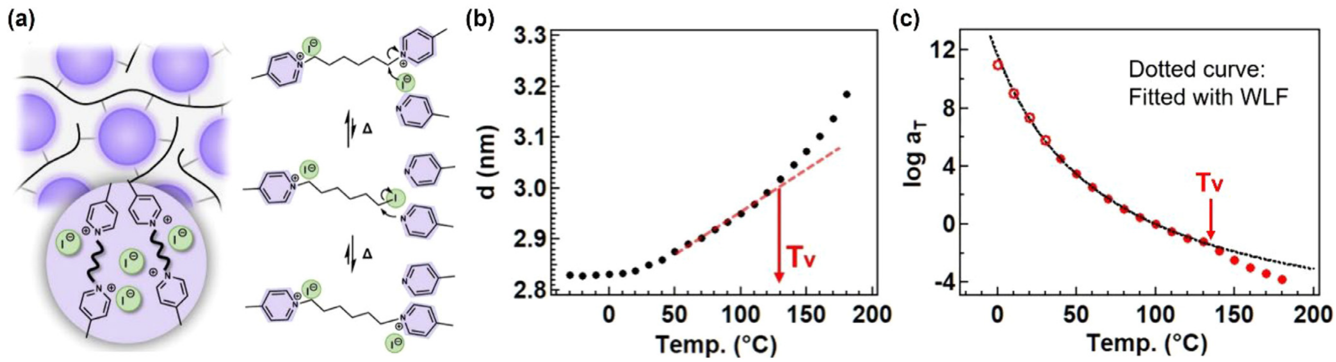


Fig. 21. (a) Molecular design of phase-separated vitrimer-like materials, including the mechanism of trans-*N*-alkylation of quaternized pyridine. Temperature dependence of (b) domain distance (d) from temperature-ramp SAXS and (c) shift factors (a_T) from SAOS master curves. [182], Copyright 2025. Adapted with permission from American Chemical Society.

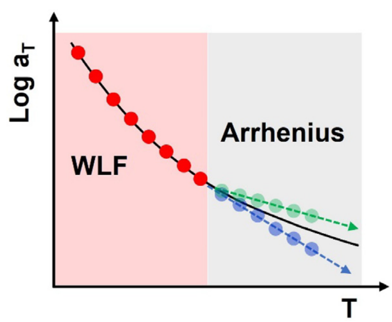


Fig. 22. Schematic illustrating the transition from WLF to Arrhenius dependence of shift factors (a_T) in SAOS master curves. Above the transition temperature, the plots deviate from WLF dependence (black line), showing either larger (blue) or lower (green) temperature dependence.

dence observed above 130 °C (Fig. 21c). Above this point, Arrhenius dependence was noted in the plots of ln a_T vs. $1/T$, with the deviation point assigned to the rheological T_v. Importantly, the rheological T_v was consistent with the structural T_v, confirming the validity of both assignments. Although the phase-separated system is uncommon, the key takeaway is the reliability of the T_v assignment protocol based on the plots of a_T.

It is important to note that, in the above system, the temperature dependence of a_T increased significantly at the transition point (i.e., at T_v) (see blue plots in Fig. 22), a trend also observed in other studies [184,185]. This is reasonable, as the molecular dynamics restricted within the frozen network are released by sufficient activation of bond exchange. However, other studies have demonstrated that at the transition point, this dependence can become weaker than the extrapolated WLF dependence (green plots in Fig. 22) [186,187]. In such cases, the overall function of shift factors is assumed to be a combination of WLF and Arrhenius behaviors; therefore, the temperature dependence should not be weaker than the original WLF dependence, even if bond exchange is fully activated above T_v. This raises an important question: what determines the direction of change in a_T at T_v if both scenarios are possible? Addressing this question is crucial for a deeper understanding of the temperature dependence of a_T or strand mobility, particularly at the transition point. The following section will provide some hints, demonstrating the current understanding of vitrimer rheology at short and long timescales.

6. Understanding of linear rheology

6.1. History of linear rheology studies for transient polymer networks

Although the mechanism governing vitrimer viscoelasticity is currently a subject of significant research focus, understanding the dynamics of transient networks is a classical polymer science problem that dates back to the previous century. During the 1940s, Stern and Tobolsky investigated the stress relaxation behavior of hydrocarbon rubbers containing disulfide cross–links [188]. Despite the covalent nature of the disulfide linkages, these rubbers exhibited exponential stress decay that followed an Arrhenius temperature dependence (Fig. 23a). Yet, upon the removal of strain

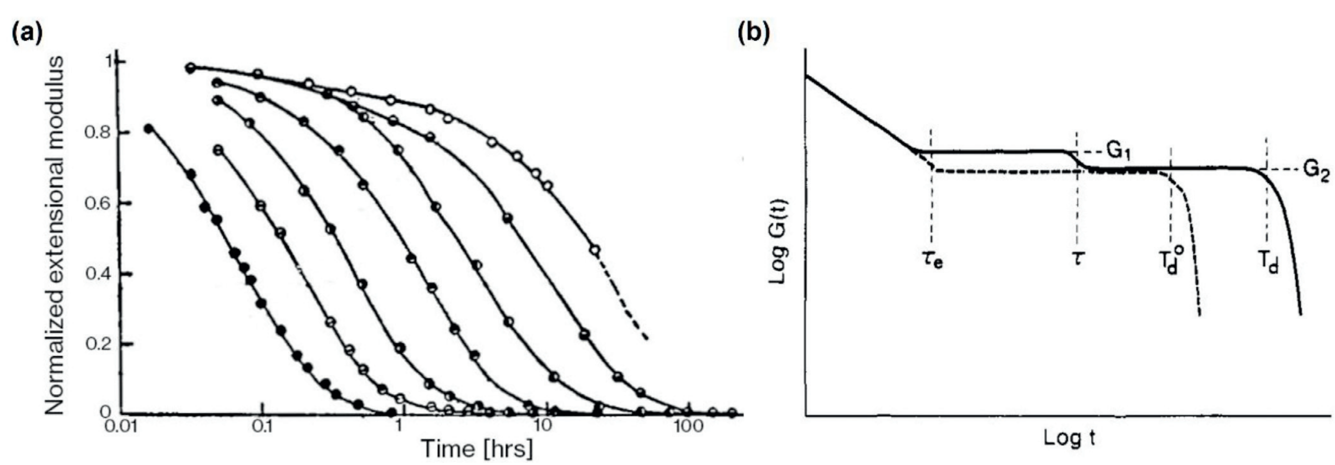


Fig. 23. History of transient polymer network rheology. (a) Stress relaxation of a polysulfide rubber at varying temperature. [188], Copyright 1946. Adapted with permission from American Chemical Society. (b) Schematic of stress relaxation for non-cross-linked chains (dashed line) and chains with reversible cross-links (solid line). [194], Copyright 1991. Adapted with permission from American Chemical Society.

they fully recovered their original modulus — demonstrating that their cross–link density remained unchanged. Building on these observations, they proposed that disulfide linkages underwent a chain–interchange process analogous to the modern mechanism of associative cross–link exchange. Their hypothesis was supported by the fact that the measured rheological activation energy (E_{rh}) matched the energy needed to break a disulfide linkage. To provide a quantitative framework for describing the rheological behavior, Green and Tobolsky formulated a kinetic theory of rubber elasticity in which the rates of network junction breaking and reformation were equal in magnitude [189]. Because this model assumes that each segment between cross-link junctions relaxes independently, the viscoelastic response is dominated by a single relaxation mode with timescale $\tau = k^{-1}$, where k is a rate constant [189]. This theory proved valuable not only for describing stress relaxation in polysulfide rubbers, but also for laying the foundation for modeling polymer entanglements [190–192].

During the early 1990s, Baxandall and Leibler, Rubinstein, and Colby independently developed the frameworks for what would eventually be known as the sticky Rouse and reptation models (Fig. 23b) [193,194]. In their models, stress dissipation along a single polymer chain requires the sequential breaking and reforming of its cross–links. Because each cross–link detaches and reattaches randomly, relaxation occurs across a spectrum of modes rather than as a single dominant process. This complex viscoelastic response is described by a bead–spring model with two primitive timescales: (I) the segmental relaxation time τ_0 and (II) cross-link lifetime τ_{XL}. τ_0 describes the time it takes a non-cross-linked backbone repeat unit to relax after the entire chain has been perturbed, while τ_{XL} represents how long a cross-link persists until it forms a new association.

6.2. Rheology of simple epoxy vitrimers

Even before the term "vitrimer" was first coined, rheology played a central role in elucidating the molecular behavior underlying these materials. Several types of rheological measurements have been employed to characterize viscoelastic behavior across different timescales: SAOS examines short timescale dynamics, stress relaxation captures intermediate timescale responses, and creep evaluates long timescale behaviors [161].

The seminal works of Leibler and coworkers established the defining linear viscoelastic signatures of epoxy-based transesterification vitrimers [15], essentially identical to the polysulfide behaviors observed by Stern and Tobolsky as shown above. Under stress relaxation, the modulus decays completely to zero, while during creep deformation the compliance increases linearly with time. The viscosity follows an Arrhenius temperature dependence with an E_{rh} that approximately matches the exchange activation energy of small molecule cross-link analogues (E_{sm}). Increasing catalyst concentration accelerates the rate of stress relaxation without altering E_{rh}, whereas changing the catalyst chemistry modulates E_{rh} [106,195].

Initial attempts at quantitatively modeling vitrimer viscoelasticity used the Green and Tobolsky kinetic theory of rubber elasticity as a foundation. For example, Long et al. derived a continuum model to describe the transient relaxation of a vitrimer under deformation [196]. The rate of cross-link breaking and reformation were both proportional to $k_0 \exp(-\frac{E_{sm}}{RT})$, where R is the gas constant, T is temperature, and k_0 is a constant. Implementing their model into three-dimensional finite element code, the nonlinear response of a vitrimer subjected to uniaxial and torsional deformations was simulated (Fig. 24a). The authors later adapted their model for evaluating light-induced relaxation and interfacial welding [197–199]. The groups of Terentjev and Vernerey used similar models to describe several epoxy vitrimer systems under linear and nonlinear deformations [149,179,200–204].

In parallel with continuum modeling efforts, early attempts were made to apply time-temperature superposition (TTS) to vitrimer viscoelastic data. In this approach, horizontal and vertical shift factors are applied to superpose the data into continuous master curves corresponding to a reference temperature T_{ref}. The temperature dependence of the horizontal shift factor, in particular, may provide insight into the underlying relaxation processes [161]. Snijkers et al., Fang et al., and Meng et al. applied TTS on epoxy vitrimers undergoing transesterification [179,181,205], while Luo et al. used it on a polyimine network (Fig. 24b and c) [206]. In all cases, TTS did not fully collapse the data across the full relaxation spectrum. When superposition was applied to the glassy transition regime, for example, it failed to superpose the rubbery plateau.

As an alternative to TTS, the linear viscoelastic response may be analyzed through conversion into a relaxation spectrum. As described in Section 4–4, in this approach the rheology data are decomposed into a discrete or continuous spectrum of relaxation modes with different strengths and timescales. Through the relaxation spectrum, major viscoelastic processes may be identified [161,207,208]. Edera et al. used the relaxation spectrum approach to decompose torsional SAOS data for an epoxy vitrimer (Fig. 24d) [170]. The resulting relaxation spectrum featured two distinct regimes with different temperature dependences: (I) fast glassy modes, i.e., the segmental motions of the epoxy backbone, and (II) slow vitrimeric modes indicative of bond exchange.

6.3. Rheology of vitrimers from commodity polymers

Vitrimer systems have achieved versatile network designs by integrating dynamic covalent linkages into commodity polyesters, polyacrylates, polymethacrylates, polyolefins, and polydienes, as documented in the previous section. This expansion of the synthetic library offered more knobs for precisely tuning rheological behavior. As a pioneering work for commodity polymers, Röttger et al. incorporated dioxaborolane linkages into poly(methyl methacrylate) (PMMA) via copolymerization [46]. Like the epoxy systems, the viscosities of the PMMA vitrimers obeyed an Arrhenius temperature dependence. The E_{rh} values, however, were larger than the E_{sm} estimated from small molecule boronic ester kinetics studies. As mentioned in Section 4.3, this discrepancy between E_{rh} and E_{sm} is seen not only for this PMMA-dioxaborolane vitrimer, but for many other systems as well.

To match the expansion of the vitrimer concept to commodity polymers, the theoretical framework for describing viscoelasticity evolved to account for the chain-like nature of the network backbone. In 2021, Ricarte and Shanbhag adapted the sticky Rouse model to evaluate the impact of associative cross-links on unentangled polymer chains [209]. Based on a generalized form of the theory, the sticky Rouse model predicted two distinct regimes, each composed of a spectrum of relaxation timescales. The fast regime corresponded to the segmental motions of the vitrimer backbone. Here, the relevant timescale is τ_0, which obeys a WLF relationship with temperature. In contrast, the slow regime represented the interplay between network strand relaxation and bond exchange. τ_{XL}, the relevant timescale for these slower relaxations, was assumed to have the functional form

$$\tau_{XL} = \tau_{XL}^0 \exp\left(\frac{E_{sm}}{RT}\right) \tag{1}$$

where τ_{XL}^0 is the bond exchange attempt time. Based on literature precedent established by the ionomer and supramolecular polymer fields [210,211], the authors assumed $\tau_{XL}^0 = \sigma \tau_0$, where σ is a tem-

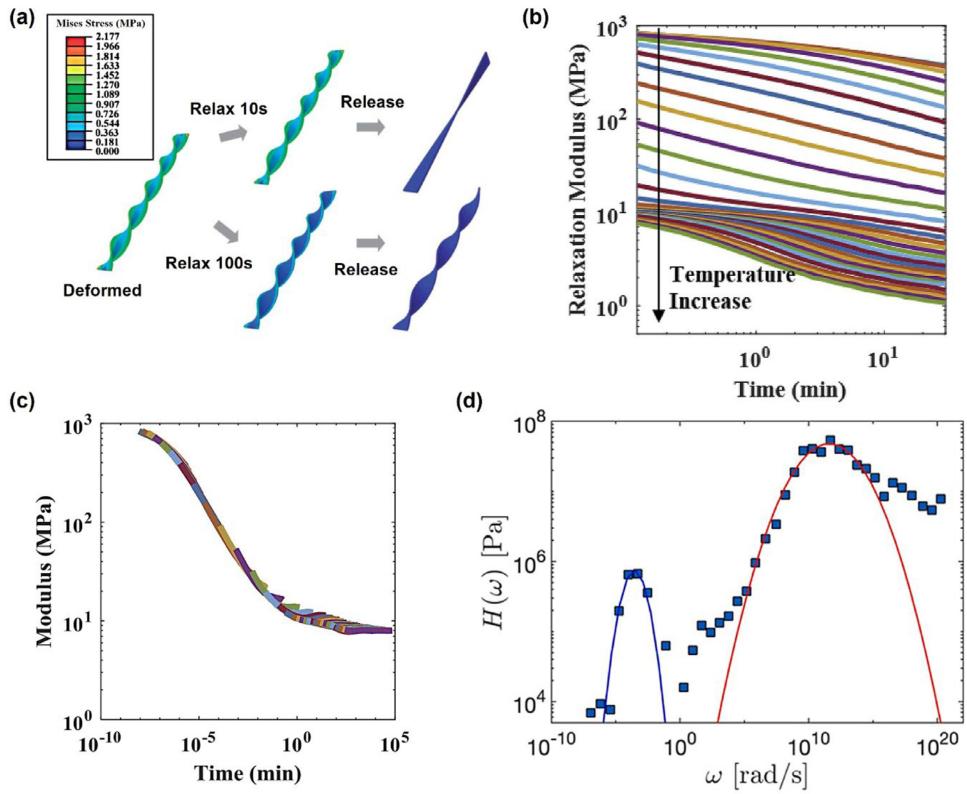


Fig. 24. Epoxy vitrimer rheology. (a) Three-dimensional stress profile simulation of an epoxy vitrimer during torsional deformation. [196], Copyright 2013. Adapted with permission from Royal Society of Chemistry. (b) Stress relaxation curves of a polyimine vitrimer without and (c) with TTS shift factors applied ($T_{ref} = 37°C$). Although the transition region at short times collapses, the rubbery–plateau regime fails to superimpose. [206], Copyright 2018. Adapted with permission from Elsevier Science Ltd. (d) Discrete and continuous relaxation spectra obtained from an epoxy vitrimer stress relaxation measurement. High- and low-ω peaks are assigned to glassy and vitrimeric modes, respectively. [170], Copyright 2024. Adapted with permission from Elsevier Science Ltd.

perature independent constant. Due to this assumption, the resulting temperature sensitivity of the slow regime is a product of the backbone WLF and bond exchange Arrhenius dependences, where $E_{rh} = E_{WLF} + E_{sm}$ and $E_{WLF} = \frac{\partial \ln \tau_0}{\partial (1/T)}$. Because τ_0 and τ_{XL} have different temperature dependences, TTS fails to superpose the entire relaxation spectrum (Fig. 25a to c).

The sticky Rouse prediction of two distinct relaxation regimes has been reproduced using simulations. Through hybrid molecular dynamics and Monte Carlo simulations, Perego et al. observed that vitrimer relaxations obeyed WLF and Arrhenius dependences at low and high temperatures, respectively (Fig. 25c) [187,212–214]. Xia et al. and Cui et al. used molecular dynamics simulation to evaluate the influence of topological defects on the slow relaxation regime [215–219]. They each found quantitative agreement between their simulations and the sticky Rouse predictions (Fig. 25d).

For experimental observations, the appearance of two distinct relaxation regimes is qualitatively consistent with rheology measurements, though it has not yielded quantitative agreement. Wu et al. used SAOS to investigate poly(hexylmethacrylate) (PHMA) vitrimers bearing dioxaborolane cross-links [186]. Across the entire angular frequency (ω) range, two distinct relaxation regimes were observed. The fast dynamics corresponded to the segmental motions of the PHMA backbone. This assignment was supported by that fact that the TTS horizontal shift factors used to superpose the high-ω regime followed a WLF temperature dependence. The slow dynamics, in contrast, had a much weaker temperature dependence. The TTS horizontal shift factors used to superpose the low-ω regime were Arrhenius in nature (Fig. 26a), producing an E_{rh} greater than the E_{sm} for dioxaborolane metathesis. Porath and Evans used both stress relaxation and SAOS to study poly(dimethylsiloxane) (PDMS) vitrimers composed of telechelic PDMS chains cross-linked by trifunctional boric acid [172]. Relaxation times obtained from stress relaxation and SAOS exhibited two distinct temperature regimes. The low- and high-temperature regimes, each following Arrhenius behavior produced activation energies of 16 and 34 kJ/mol, respectively (Fig. 26b). Ricarte et al. employed a combination of SAOS, creep, and stress relaxation measurements to study the linear viscoelasticity of polybutadiene (PB) vitrimers with dioxaborolane cross-links [220]. Rheology measurements were performed at temperatures between 80 and 160 °C, well above the PB vitrimer glass transition temperature. SAOS and creep measurements produced two separate sets of TTS horizontal shift factors (a_{SAOS} and a_{creep}, Fig. 26c). Application of a_{SAOS} and a_{creep} on stress relaxation data collapsed the short and long time regimes, respectively. Arrhenius analyses on a_{SAOS} produced activations energies that were equal to the E_{WLF} for PB, supporting the argument that the fast relaxations represented backbone segmental motions. Arrhenius analyses on a_{creep}, however, produced activation energies much larger than sticky Rouse predictions.

The observation of two distinct fast and slow relaxations was not limited to the system using dioxaborolane exchange. Barzycki et al. evaluated polystyrene (PS) vitrimers with dynamic imine linkages based on multiple rheological measurements [221]. For SAOS, high-ω shift factors displayed a WLF dependence that exactly matched the behavior of non-cross-linked PS, serving as additional evidence that the fast relaxations corresponded to segmental motions. However, low-ω shift factors exhibited Arrhenius behavior, with a temperature dependence significantly weaker than that predicted by the sticky Rouse model and even weaker than

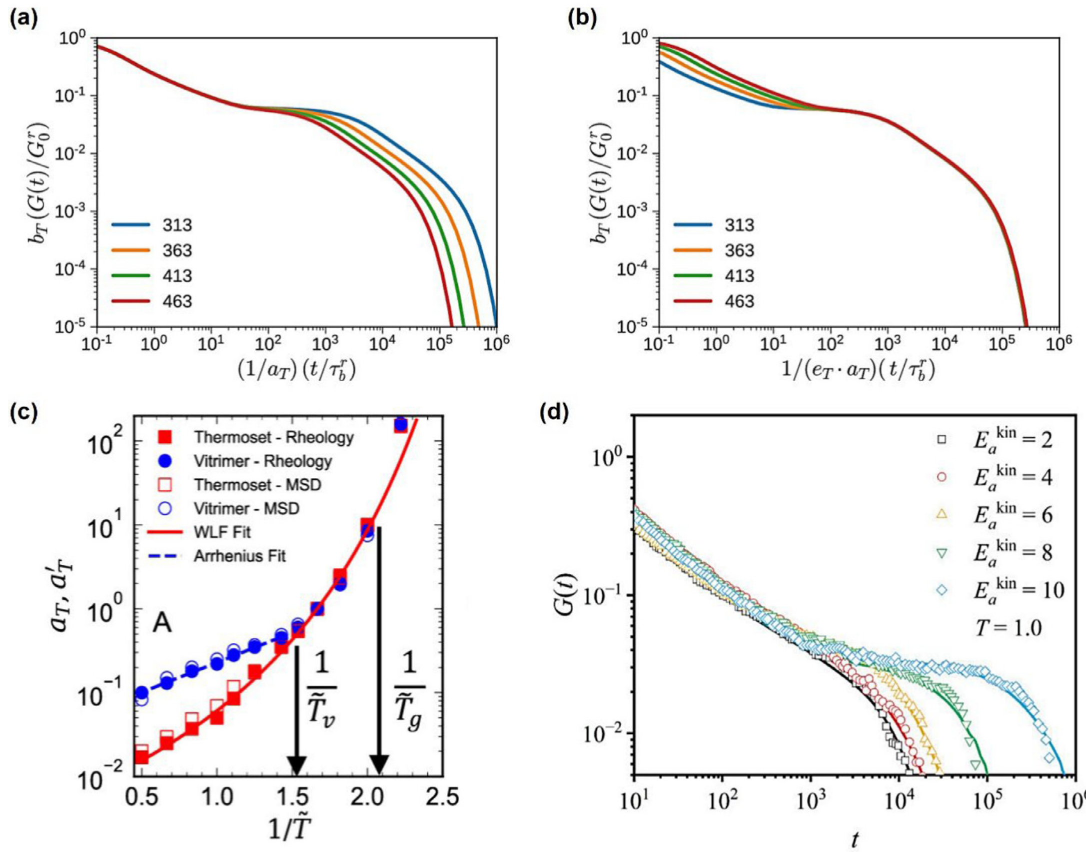


Fig. 25. Simulations of vitrimer rheology. Sticky Rouse pseudo-master curves of unentangled vitrimer stress relaxation with (a) short time and (b) long time horizontal shift factors applied. [209], Copyright 2021. Adapted with permission from American Chemical Society. (d) Hybrid Monte Carlo and molecular dynamics horizontal shift factors for polymer networks with permanent cross-links (thermoset) and vitrimers. [187], Copyright 2022. Adapted with permission from American Chemical Society. (d) Molecular dynamics of vitrimers with varying bond exchange activation energy. Open circles are the simulated data. Solid lines are generalized Maxwell model fits. [216], Copyright 2023. Adapted with permission from American Chemical Society.

the segmental motions themselves (Fig. 26d). Decomposition of the rheological data into relaxation spectra produced a similar conclusion (Fig. 26e). In a similar manner, Ge and Evans found that acrylic and methacrylic vitrimers with imine cross-links displayed fast WLF-like segmental relaxations and slow Arrhenius dynamics [131]. Note that, in both of these imine cross-link systems, E_{rh} was weaker than E_{WLF}.

6.4. Open questions in vitrimer rheology

Theory, simulations, and experiments have revealed common linear viscoelastic behaviors across many vitrimer systems. In particular, the rheological response exhibits two distinct relaxation regimes. Considerable experimental evidence shows that the fast regime reflects the segmental motion of the polymer backbone. In contrast, the mechanism governing the slow relaxation regime remains unclear. E_{rh} is generally greater than E_{sm}, indicating that local bond exchange is not sole factor that determines the long time behavior. At the same time, E_{rh} can be greater or less than E_{WLF}, somewhat obscuring the relationship between the backbone segmental motions and slow regime. One potential key to resolving the underlying mechanism lies in identifying the critical factors that control τ_{XL}: E_{sm} and τ^0_{XL} (see Eq. (1)).

E_{sm} represents the energy barrier for bond exchange. Typically, experimental determination of E_{sm} involves tracking the exchange kinetics of small molecule cross-link analogues in either solvent or bulk [15,46,106,195]. However, transferring the cross-links to a polymeric matrix potentially alters the solvation environment of the reaction. The matrix polarity, for example, may affect the stability of any potential cross-link intermediates, thereby modifying both the local E_{sm} and macroscopic E_{rh} [123,221,222]. Density functional theory calculations may offer a pathway to isolate how E_{sm} changes as a function of the vitrimer matrix [223–225]. τ^0_{XL} represents the bond exchange attempt time, essentially the period it takes the reactive units of the bond exchange to find each other. The physical processes that dictate this timescale, however, remain unclear. For other types of transient polymer networks, e.g., ionomers and supramolecular polymers, τ^0_{XL} is assumed to be directly equal to τ_0. Recent work, however, suggests that this assumption does not hold true for vitrimers [131,160,221].

One possible explanation is that τ^0_{XL} is controlled by the diffusion of the cross-linker within the vitrimer matrix. Barzycki et al. proposed this hypothesis to rationalize the approximately 100 kJ/mol difference between E_{rh} and E_{sm} for PS vitrimers with imine linkages [221]. Using the free volume theory, they estimated that the diffusion activation energy of residual cross-linker within the PS matrix is on the order of 100 kJ/mol, providing a plausible rationale for the E_{rh} and E_{sm} difference (Fig. 27a). Ge and Evans also acknowledged that cross-linker diffusion may be a governing factor during the relaxation of acrylic and methacrylic vitrimers [131]. Some simulations and experiment studies have explored the relationship between backbone segmental motions and small molecule diffusion within vitrimer matrices [226–229]. Yet, deeper investigations are needed to clarify the significance of cross-linker mobility on τ_{XL}.

Eq. (1) also may be revised to account for additional physical processes. The Renormalized Bond Lifetime Model (RBLM), originally developed for dynamic networks that undergo dissociative

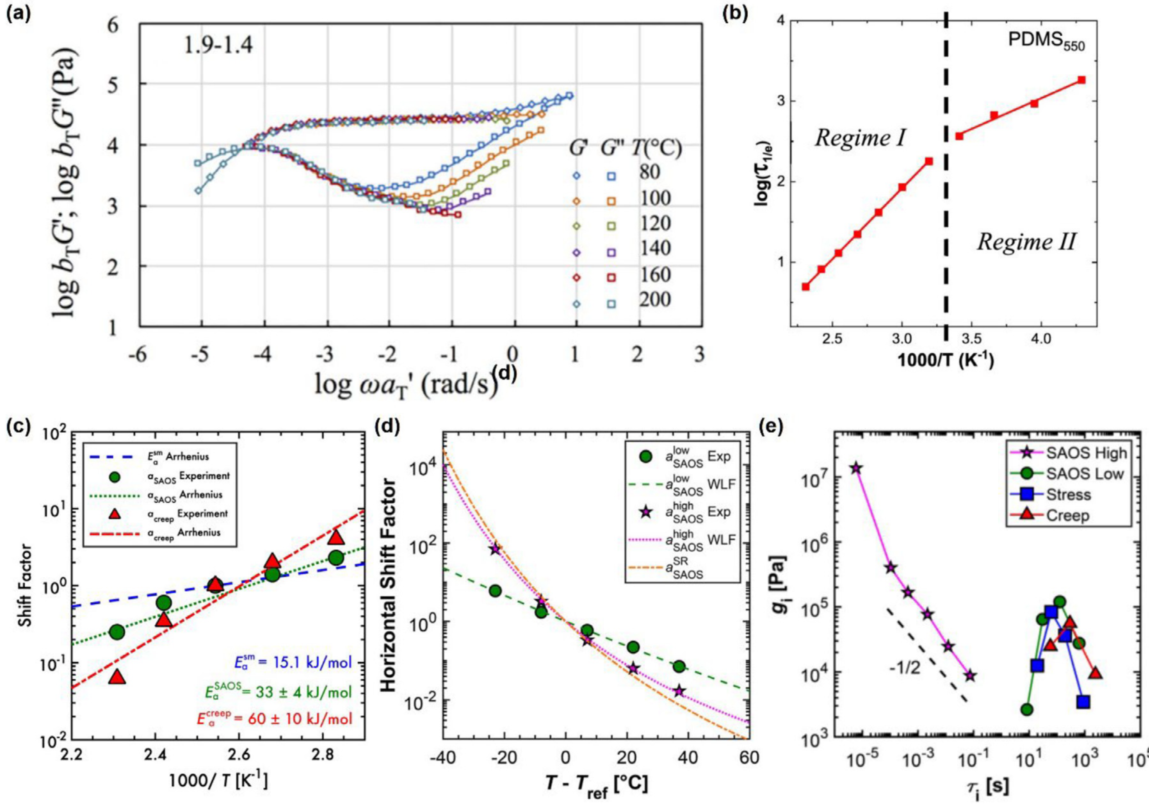


Fig. 26. Experimental measurements of vitrimer rheology. (a) SAOS pseudo-master curve of a PHMA vitrimer with dioxaborolane cross-links. Horizontal shift factors were applied to collapse low-ω regime at ($T_{ref} = 80\,^{\circ}$C). [186], Copyright 2020. Adapted with permission from American Chemical Society. (b) Relaxation time temperature dependence for PDMS vitrimers with boric acid cross-links. [172], Copyright 2021. Adapted with permission from American Chemical Society. (c) Horizontal shift factors for PB vitrimers with dioxaborolane cross-links at ($T_{ref} = 120\,^{\circ}$C). SAOS and creep shift factors probe fast and slow relaxations, respectively. Adapted with permission from [220]. Copyright (2023) American Chemical Society. (d) Horizontal shift factors ($T_{ref} = T_g + 50\,^{\circ}$C) and (e) discrete relaxation spectra for PS vitrimers with imine cross-links. [221], Copyright 2025. Adapted with permission from American Chemical Society.

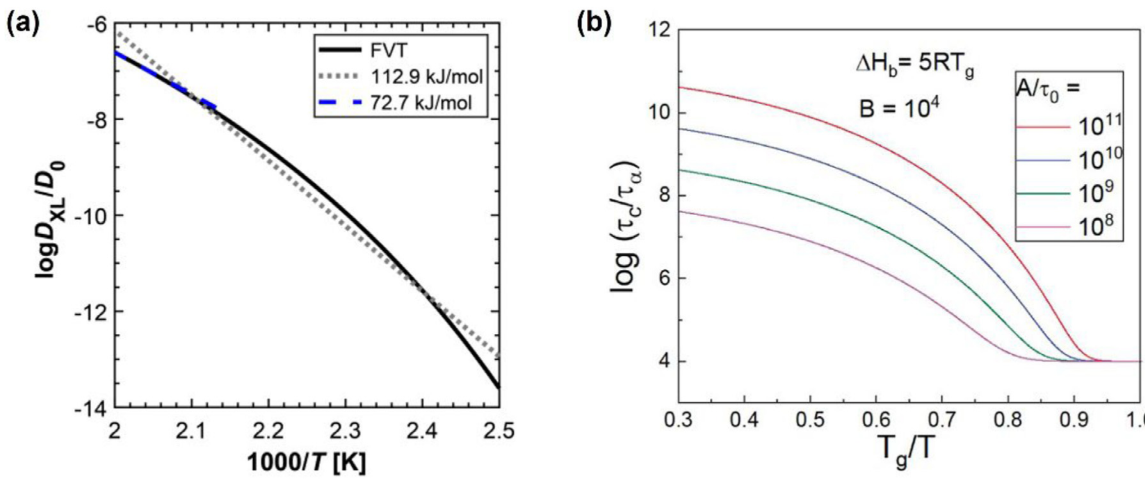


Fig. 27. (a) Estimate of 1,6-hexanediamine cross-linker diffusion coefficient (D_{XL}) in a PS matrix as a function of temperature, based on the free volume theory (FVT). Gray dotted and blue dashed lines represent Arrhenius analyses over different temperature ranges: the gray dotted line for the entire temperature range and the blue dashed line for the limited high temperature range. [221], Copyright 2025. Adapted with permission from American Chemical Society. (b) Simulations of the cross-link lifetime based on Eq. (3). In the Y-axis, τ_c and τ_α represents the terminal relaxation timescale and the segmental relaxation timescale, respectively; the axis is thus an indicator of τ_{XL}. [131], Copyright 2025. Adapted with permission from American Chemical Society.

bond exchange, modifies τ_{XL} to account for cross-link reassociations that do not contribute to network relaxation

$$\tau_{XL} = J\tau_{XL}^{0} \exp\left(\frac{E_{sm}}{RT}\right) + \tau_{open} \tag{2}$$

where J is the number of times a cross-linkable group reassociates with an old partner and τ_{open} is the time it takes an unbound group to find a new partner [230–233]. Fractional Brownian motion simulations suggest that J can be on the order of 100, indicating that cross-links typically undergo several reassociations before finding a new partner [234]. While the RBLM predicts that reassociations increase the terminal relaxation time, it predicts a smaller difference between E_{rh} and E_{sm} than Eq. (1) [220].

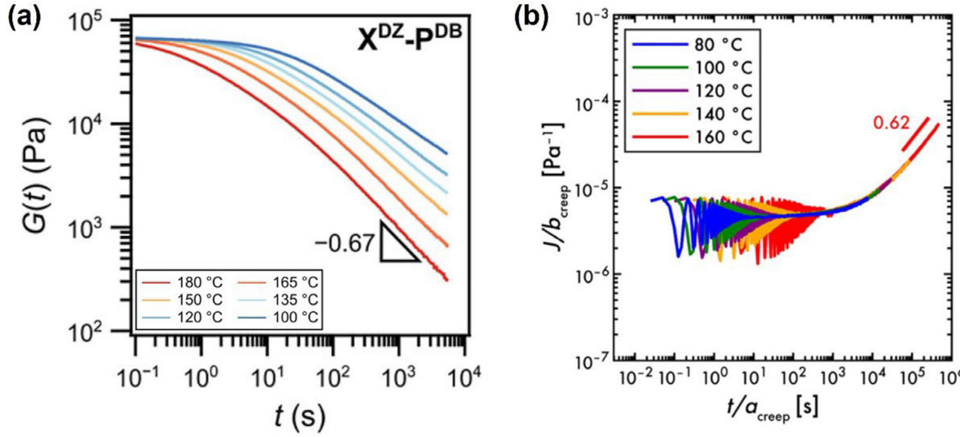


Fig. 28. (a) Stress relaxation of methacrylic vitrimer with dioxazaborocane cross-links. Power law behavior is observed over several orders of magnitude in time. [238], Copyright 2025. Adapted with permission from American Chemical Society. (b) Creep of PB vitrimers with dioxaborolane cross-links at ($T_{ref} = 120\,^\circ$C). After more than 10^5 s, the vitrimer still does not attain the terminal relaxation power law scaling of 1. [220], Copyright 2023. Adapted with permission from American Chemical Society.

For vitrimers, recent studies propose a τ_{XL} expression that is directly inspired by the RBLM [131,160,225,235], written as,

$$\tau_{XL} = A \exp\left(\frac{\Delta H_b}{RT}\right) + B\tau_0 \tag{3}$$

The first term in Eq. (3) represents the timescale for local bond exchange, where E_{sm} is replaced with an explicit enthalpic energy barrier for bond exchange (ΔH_b). The pre-exponential factor A essentially sets the frequency for bond exchange attempts. While A likely depends on the local concentration of reactive groups [162], Rukmani et al. propose that it also hinges on the probability that the reactive groups attain the correct orientation needed for bond exchange to occur [236]. Borrowing the knowledge of heterogeneous catalyst from literature, they describe this orientation criterion using the steric factor. The second term in Eq. (3) describes the time it takes an associative cross-link to encounter a reactive partner, with the parameter B modulating its overall significance. Due to the presence of τ_0, Eq. (3) implicitly assumes that this process depends on the backbone segmental motions [225,234]. Ge and Evans then evaluated the significance of A, ΔH_b, and B on τ_{XL} (Fig. 27b) [131].

One additional characteristic of many vitrimer systems that has recently come to light is the lack of terminal relaxation features in the slow regime. Specifically, terminal relaxation is indicated by the onset of specific power scalings: (I) $G' \sim \omega^2$ and $G'' \sim \omega^1$ in SAOS and (II) $J(t) \sim t^1$ in creep where $J(t)$ is the creep compliance. In stress relaxation, terminal relaxation corresponds to an exponential decay in $G(t)$ [161]. These features are sometimes observed for vitrimers with telechelic cross-links [172,237]. Yet, many vitrimers with pendant cross-links do not exhibit the characteristic scalings, even after very long period of time at temperatures well above T_g and T_v. Instead, they seem to express non-terminal power law behavior over several orders of magnitude in time (Fig. 28a and b) [131,220,221,225,238]. This behavior is reminiscent of critical gels, in which the network is at the percolation threshold [239,240]. Further investigation is needed to determine if this extended power law regime is just due to delayed terminal relaxation, or if it is an indication of an alternative type of relaxation process [241,242].

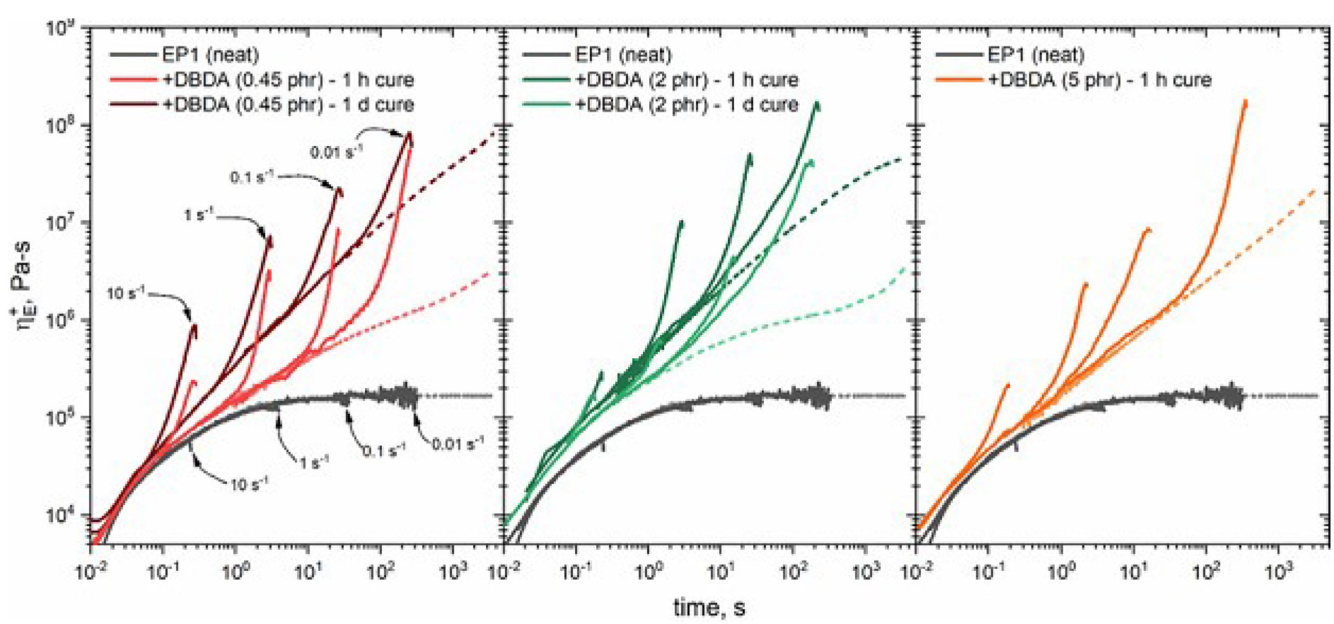


Fig. 29. Start-up extensional flow viscosity (η_E^+) measurements for ethylene-propylene (EP) vitrimers with dioxaborolane cross-links. Dashed lines depict the linear viscoelastic envelopes, while solid lines are the experimentally measured data. [57], Copyright 2024. Adapted with permission from American Chemical Society.

Although significant progress has been made in understanding vitrimer linear rheology, their nonlinear behavior remains largely unexplored. This knowledge gap is critical because the industrial and commercial viability of vitrimers hinges on their performance under large strain and shear processing conditions. Initial nonlinear extensional rheology measurements by López-Barrón et al. and Zhang et al. demonstrate that some vitrimers may undergo strain hardening (Fig. 29) [57,168,243]. Nevertheless, there is still significant opportunity to understand the high strain flow behavior through experiment, simulations, and theory. These fundamental knowledge could further expand the potential of vitrimers for various processing techniques.

7. Conclusions

In this review, we have highlighted recent significant developments and identified key open questions for vitrimers. Since the pioneering work of L. Leibler et al., diverse vitrimer designs have been explored, incorporating various bond exchange mechanisms within different polymer frameworks. For practical applications, overcoming high viscosity remains a critical challenge for manufacturing processes that require precise shaping. To address this, there have been studies demonstrating the trade-off between mechanical properties and processability. For this issue, post-molding curing systems may be attractive. However, the adoption of injection and extrusion molding techniques for cured vitrimers remains limited, which also hampers the mechanical recycling of vitrimer-based products. Therefore, exploring additional practical functionalities is of great importance. Progress in characterizing vitrimers—particularly their stress relaxation behavior, topology freezing temperatures, and fundamental rheological properties—has been achieved through combined experimental, simulation, and theoretical approaches. These intrinsic properties are closely linked to processing and performance (e.g., recyclability and healabiltiy), underscoring their significance for practical applications. As the understanding of vitrimers deepens, new and intriguing questions inevitably emerge. The open questions outlined in this review aim to guide future research and facilitate the continued advancement of vitrimer science.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Mikihiro Hayashi: Writing – original draft, Validation, Investigation, Writing – review & editing, Visualization, Supervision, Conceptualization. **Ralm G. Ricarte:** Writing – original draft, Validation, Conceptualization, Writing – review & editing, Visualization, Investigation.

Data availability

The authors do not have permission to share data.

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